

Interplay of YEATS2 and GCDH regulates histone crotonylation and drives EMT in head and neck cancer

Deepak Pant, Parik Kakani, Rushikesh Joshi, Abin Sabu, Shruti Agrawal, Atul Samaiya, Sanjeev Shukla 

Department of Biological Sciences, Indian Institute of Science Education and Research Bhopal, Bhopal, India • Department of Pathology, Bansal Hospital, Bhopal, India • Department of Surgical Oncology, Bansal Hospital, Bhopal, India

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These **useful** findings assigned a novel functional implication of histone acylation, crotonylation. Mechanistic insights have been provided in great detail regarding the role of the YEATS2-GCDH axis in modulating epithelial-to-mesenchymal transition (EMT) in head and neck cancer, and overall the strength of evidence is **solid**.

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Abstract

The regulation of gene expression is an integral cellular process orchestrated by epigenetic marks like histone modifications. Perturbations in the activity or abundance of epigenetic factors can lead to tumorigenesis. Remarkably, several metabolites influence the epigenetic landscape in cells. Here, we investigated the interplay between a highly expressed epigenetic factor, YEATS2, and a metabolic enzyme, GCDH, in regulating epithelial-to-mesenchymal transition in head and neck cancer. We report that the histone reader YEATS2 is responsible for increased invasive potential in head and neck cancer in an SP1-dependent manner. YEATS2 functions by maintaining histone crotonylation, and its abrogation leads to a global decrease in the H3K27cr mark. Mechanistically, we report that YEATS2 maintains high promoter H3K27cr levels by assisting in the recruitment of crotonyltransferase p300 at the promoter of the EMT-promoting gene *SPARC*. Further, we found that the addition of the H3K27cr mark is also dependent on the crotonyl-CoA-producing enzyme GCDH. Overall, we describe a novel mechanism of interplay between epigenetics and metabolism in head and neck tumorigenesis, which results in the enhanced expression of EMT-related genes in a histone crotonylation-dependent manner.

Introduction

Head and neck cancer is an umbrella term associated with a number of cancers of the oral cavity and pharynx. Like other solid tumors of epithelial origin, head and neck cancer (HNC) involves a multi-step progression towards malignancy, of which transition from an epithelial to a mesenchymal phenotype is a pivotal event ¹. This epithelial-to-mesenchymal transition (EMT) involves dramatic changes like loss of cell polarity and acquisition of migratory and invasive properties in the tumor cells ². Underlying these remarkable external changes in tumor cells, are equally notable changes in gene expression and signalling pathways. Moreover, widespread alteration in the gene expression program of cancer cells undergoing EMT can be a result of either altered activity or the change in abundance of several epigenetic factors. Tumor-specific epigenomic changes thus allow HNC cells to tweak the expression levels of multiple genes at once, thereby leading to a drastic transformation in phenotype during metastasis. These large-scale changes are due to the rapid and global nature of epigenetic alterations ³.

Similar to epigenetic changes, metabolic rewiring is another important hallmark of cancer cells capable of carrying out vast tumor-promoting changes rapidly. Numerous reports in the past have documented instances of loss-of-function mutations or changes in the gene expression of enzymes involved in key metabolic processes, such as glycolysis or the citric acid cycle, to promote oncogenesis ⁴. However, the discovery of non-canonical functions of several enzymes and their atypical localization in the nucleus has intrigued cancer researchers of late. Metabolic enzymes are now believed to play a more direct role in influencing oncogene expression via crosstalk with the epigenome. One such phenomenon is the role of metabolites produced as a result of aberrant metabolism in acting as substrates for epigenetic factors. A ground breaking study in this regard by Wellen et al. reported the nuclear localization of a mitochondria-resident protein ACLY/ACL (adenosine triphosphate–citrate lyase). Nuclear pool of ACLY was found to affect histone acetylation directly by producing acetyl-coenzyme A (CoA) in colon carcinoma cells ⁵. Moreover, another study reported a direct interaction involving α -ketoglutarate dehydrogenase (α -KGDH) and a histone acetyltransferase leading to the production of localized succinyl-CoA in nucleus, which in turn led to histone succinylation and enhanced expression of genes related to tumor cell proliferation ⁶. M2 isoform of pyruvate kinase (PKM2) is another metabolic enzyme known to exhibit a non-canonical function by directly affecting gene transcription. We have recently reported the role of nuclear PKM2-HIF1 α -p300 axis in maintaining high levels of H3K9ac marks at the promoter of *PFKFB3*, which thereby leads to its increased expression in hypoxic breast cancer cells ⁷. The reports stated herein, along with others ⁸, delineate the role of the nexus between aberrant metabolism and the epigenome in regulating various aspects of tumorigenesis in a wide variety of cancers. However, the significance of this partnership in regulating EMT, which is one of the most vital events in cancer, remains largely unexplored.

In this study, we have made an attempt to dissect the role of previously unknown link between epigenetic factors and metabolism in promoting EMT in head and neck cancer. To explore tumor-promoting epigenetic factors in HNC we analyzed large gene expression dataset and narrowed down our search to YEATS2. This gene was found to be overexpressed in HNC, had significant correlation with EMT, and also showed association with poor patient prognosis. YEATS2 (YEATS-domain containing protein 2) is a histone reader protein capable of recognizing epigenetic marks such as histone acetylation and crotonylation ⁹. Through RNA-seq, it was revealed that silencing of YEATS2 led to a decrease in the expression of multiple EMT-associated genes. Further, the transcription factor SP1 was found to be responsible for YEATS2 expression in HNC. While exploring the relationship of YEATS2 with cancer cell metabolism, we found that YEATS2 is co-expressed with crotonyl-CoA producing GCDH (glutaryl-CoA dehydrogenase) in HNC, and the downregulation of either of these two genes led to a decrease in H3K27cr levels in HNC cells. The H3K27cr mark has been previously reported to activate the transcription of pro-tumorigenic *ETS1*

in colorectal cancer ¹⁰. Corroborating with the previous finding, high H3K27cr levels at the promoter region was associated with an increase in the expression of EMT-related *SPARC* gene. We also demonstrated that YEATS2 interacts with the epigenetic writer p300 to recruit it to the *SPARC* promoter, facilitating the deposition of H3K27cr marks. Subsequent investigation revealed that YEATS2 and GCDH worked synergistically in maintaining higher levels of H3K27cr at the promoter of *SPARC*, thereby causing an increase in its expression. Finally, H3K27cr ChIP-seq in YEATS2-abrogated cells uncovered the role of YEATS2 in maintaining global histone crotonylation on the promoter of a large number of genes, many of those genes being associated with EMT. Altogether, we have elucidated a novel link that unifies epigenetic regulation and metabolic rewiring in contributing toward the enhancement of EMT in head and neck cancer.

Results

YEATS2 is overexpressed in head and neck cancer

In order to study cancer-associated epigenetic factors, we analysed publicly available The Cancer Genome Atlas (TCGA) mRNA expression dataset. The analysis aimed at obtaining genes coding for epigenetic factors that are overexpressed in HNC, and are also associated significantly to overall patient survival (**Figure 1A**). The analysis resulted in a list of 10 epigenetic factors that are upregulated in HNC and whose expression levels are associated with poor overall patient survival (**Figure 1B**, Supplementary Table 1). Out of these 10 epigenetic factors, *YEATS2*, *RUVBL1* and, *MRGBP* were not explored rigorously in HNC prior to our study ^{11–17}. The expression of these three epigenetic factors was thus evaluated at mRNA level using reverse-transcription quantitative real-time PCR (RT-qPCR) in pairs of 8 head and neck normal and cancer tissues collected as a part of this study. We found that *YEATS2*, that encodes for a histone reader protein, was the only gene that was significantly upregulated in cancer as compared to normal tissues (**Figure 1C–D**, Figure 1—figure supplement 1A–B). *YEATS2* was also found to be upregulated in publicly available HNC microarray datasets GSE30784 and GSE9844 (**Figure 1E–F**). The association of *YEATS2* with EMT was then established by significant positive correlation observed between its expression and the expression of genes included in an EMT gene signature in TCGA data (Figure 1—figure supplement 1C). Moreover, the expression of *YEATS2* was upregulated in higher tumor grades (grades 2–4) as compared to grade 1 tumors (Figure 1—figure supplement 1D). Additionally, on performing gene set enrichment analysis (GSEA) using TCGA data, we found that *YEATS2* expression is significantly correlated with the expression of metastasis-associated genes (Figure 1—figure supplement 1E). On further investigation using immunoblotting, we found that the expression of *YEATS2* protein was increased in a set of 23 head and neck cancer vs. matched normal tissues (**Figure 1G**, Figure 1—figure supplement 1F).

In order to study the role of *YEATS2* in the progression of HNC, we performed RNA-seq in *YEATS2*-silenced BICR10 cells. The shRNA-mediated silencing of *YEATS2* led to the downregulation of 2,494 genes, whereas 2,073 genes were found to be upregulated (**Figure 1H**, Supplementary Table 2). Overrepresentation analysis revealed that hallmark gene sets from MSigDb (Molecular signature database) like OXIDATIVE PHOSPHORYLATION and EPITHELIAL MESENCHYMAL TRANSITION were some of the top pathways enriched among the downregulated genes (**Figure 1I**). On the other hand, pathways like TNFA SIGNALING VIA NFKB and G2M CHECKPOINT were enriched among the upregulated genes (Figure 1—figure supplement 1G). These set of results hint towards the role of *YEATS2* in being one of the essential epigenetic factors responsible for HNC tumorigenesis, mediated via global control of EMT genes' expression.

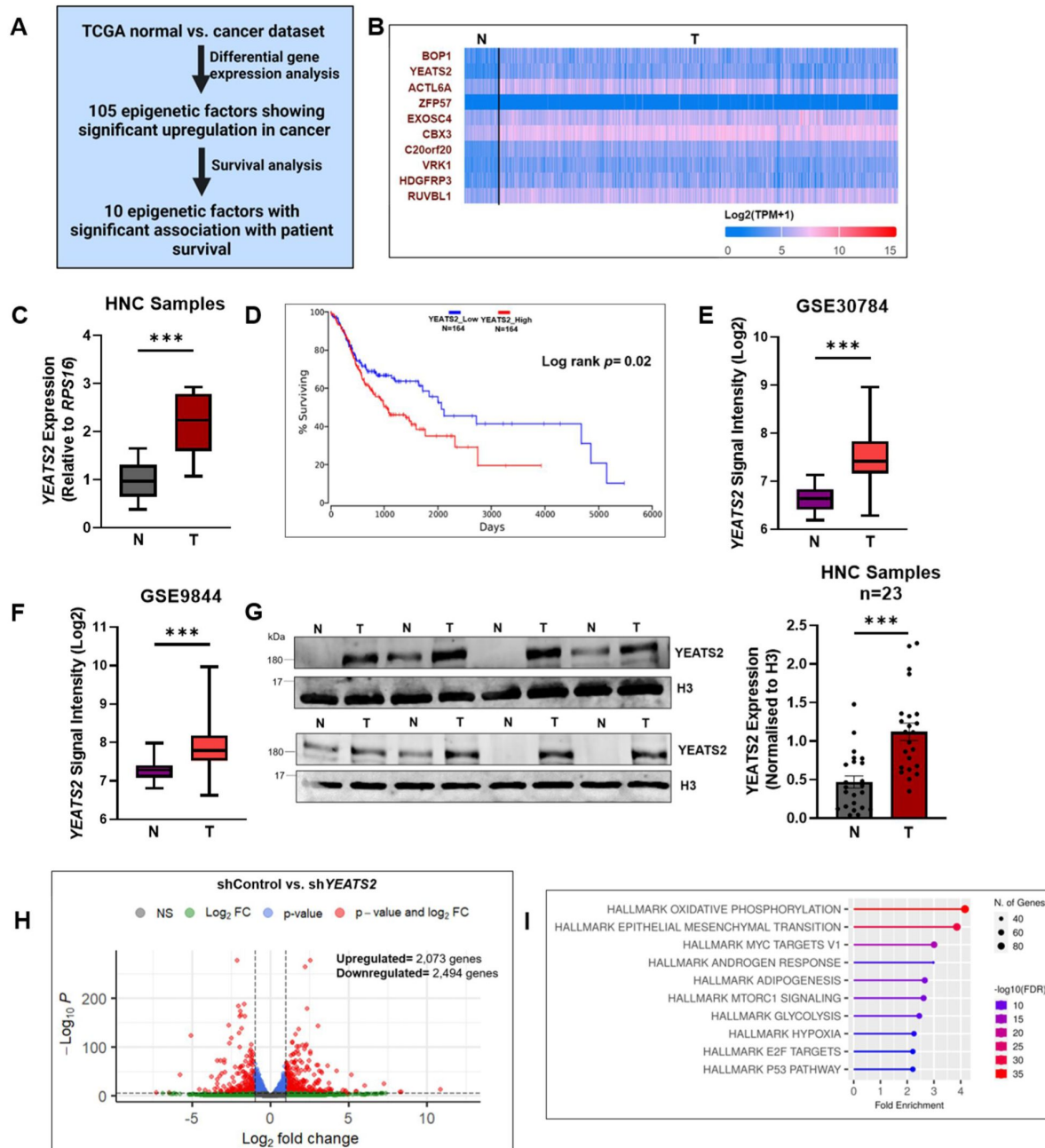


Figure 1.

YEATS2 is upregulated in head and neck cancer.

(A) Computational strategy used to find epigenetic factors with poor patient prognosis overexpressed in HNC. (B) Heatmap depicting the TCGA mRNA expression of top 10 epigenetic factors shortlisted using strategy mentioned in (A). (C) RT-qPCR result showing mRNA expression of *YEATS2* in HNC samples (n=8). (D) Kaplan-Meier curve of *YEATS2* showing association with overall head and neck cancer patient survival. (E-F) Gene expression profile of *YEATS2* in publicly available HNC microarray datasets, GSE30784 (E) and GSE9844 (F). (G) Immunoblot showing protein levels of *YEATS2* in nuclear protein lysates extracted from HNC tissues (n=23). (H) Volcano plot of differentially expressed genes in RNA-seq analysis of shControl vs. sh*YEATS2* in BICR10 cells. (I) Results of overrepresentation analysis of genes significantly downregulated in shControl vs. sh*YEATS2* RNA-seq data. Error bars, min to max; two-tailed t test, *** $p < 0.001$, N-Normal, T-Tumor.

YEATS2 is associated with increased invasion in head and neck cancer cells

Since YEATS2 was found to influence the expression of EMT-related genes globally in our RNA-seq data, we examined the expression of EMT markers in shControl vs. shYEATS2 BICR10 and SCC9 cells using immunoblotting and found that Twist1, N-cadherin and, Vimentin showed a decrease upon YEATS2 knockdown (**Figure 2A**, Figure 2—figure supplement 1A). Also, the number of invading cells significantly reduced in Matrigel invasion assay on YEATS2 downregulation, indicating that YEATS2 could be playing a role in the maintenance of EMT phenotype in HNC cells (**Figure 2B**, Figure 2—figure supplement 1B). Conversely, the expression of Twist1 was increased and there was significant upregulation in invasive capacity of cells when BICR10 and SCC9 cells were transfected with YEATS2-overexpression construct (**Figure 2C**, Figure 2—figure supplement 1C-D). Wound healing assay was performed to assess the cell migration property of HNC cells and it was found that YEATS2-silenced cells migrated slower than control cells (**Figure 2D**, Figure 2—figure supplement 1E). On the other hand, overexpressing YEATS2 in BICR10 and SCC9 cells resulted in faster wound healing as compared to control cells (**Figure 2E**, Figure 2—figure supplement 1F). A 3D invasion assay was also performed to check the ability of tumor spheres to invade a collagen matrix in YEATS2-knockdown and -overexpressed conditions. It was observed that a reduced number of cells invaded the collagen matrix when YEATS2 was downregulated. The opposite pattern was observed in both the cell lines when YEATS2 was overexpressed (**Figure 2F-G**, Figure 2—figure supplement 1G-H). All of the above observations led us to the conclusion that YEATS2 is one of the important epigenetic factors responsible for conferring head and neck cancer cells with invasive properties.

SP1 regulates the expression of YEATS2 in head and neck cancer

In order to investigate the molecular mechanism behind the upregulated expression of YEATS2 in HNC, YEATS2-promoter fragments of variable lengths were cloned upstream of Firefly luciferase reporter gene in the pGL3-Basic vector. Luciferase assay was performed in two different HNC cell lines, BICR10 and SCC9. A significant decrease in the luciferase activity was observed for shorter promoter-deletion construct (Luc-311) compared to the full-length construct (Luc-508) (**Figure 3A**, Figure 3—figure supplement 1A). Transcription factor binding-site analysis was then performed with the sequence unique to Luc-508, the region potentially responsible for transcriptional regulation of YEATS2. This analysis revealed the presence of multiple consensus binding sites of SP1 and KLF5 transcription factors. Expression profiles of these transcription factors in the TCGA mRNA expression dataset for HNC showed that unlike *KLF5*, *SP1* was significantly upregulated in tumor vs. normal patient data (Figure 3—figure supplement 1B-C). Furthermore, shRNA-mediated silencing of SP1 led to decreased expression of YEATS2 in the HNC cell lines (**Figure 3B-C**, Figure 3—figure supplement 1D-E). Further, on performing SP1-ChIP assay, we found significant enrichment of SP1 on YEATS2 promoter when compared with control IP in both the cell lines (**Figure 3D**, Figure 3—figure supplement 1F). Next, we also performed luciferase assay using Luc-508 construct in SP1-knockdown condition, and found a decreased luciferase activity as compared to control (**Figure 3E**). To further validate the direct involvement of SP1 in the transcriptional regulation of YEATS2, 3 of the 4 SP1 consensus binding sites in the full-length YEATS2 Luc-508 were mutated using site-directed mutagenesis. As expected, after performing luciferase assay, nearly 50% decrease in luciferase activity was observed for the mutant promoter construct, compared to its wild-type counterpart (**Figure 3F**, Figure 3—figure supplement 1G). We then looked into the status of invasion in shSP1 cells and found that the EMT-marker Twist1 was reduced in these cells. However, the ectopic expression of YEATS2 in shSP1 cells was able to rescue Twist1 expression partially (**Figure 3G**). We then performed invasion assay to check the effect of YEATS2 overexpression in the background of SP1-knockdown, and

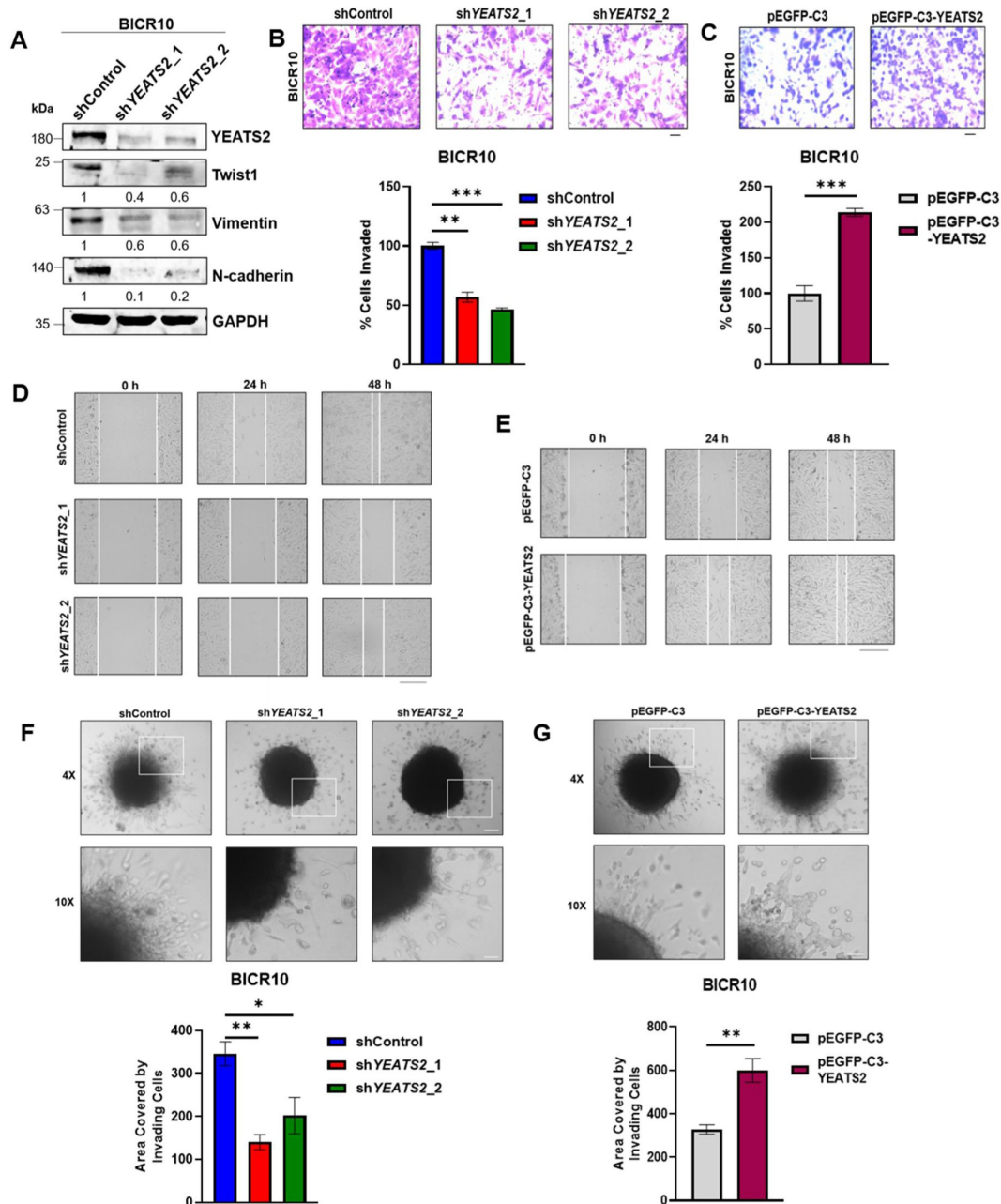


Figure 2.

YEATS2 drives EMT in head and neck cancer cells.

(A) Immunoblot showing the expression levels of various EMT factors upon YEATS2 knockdown in BICR10. (B-C) Results of invasion assay with quantification (below), after knockdown (B) or overexpression (C) of YEATS2 in BICR10 (Scale bar, 200 μ m). (D-E) Wound healing assay performed after knockdown (D) or overexpression (E) of YEATS2 in BICR10 (Scale bar, 275 μ m). (F-G) Results of 3D invasion assay showing change in invasive potential of BICR10 cells in collagen matrix after silencing (F) or overexpression (G) of YEATS2 (quantification shown below). (Scale bar: 4X, 200 μ m; 10X, 50 μ m) Error bars, mean $\pm$ SEM; two-tailed t test, * p < 0.05, ** p < 0.01, *** p < 0.001, n = 3 biological replicates.

found that there was a decrease in invasion in sh*SP1* cells, which was rescued by overexpression of YEATS2 (**Figure 3H** [↗](#), Figure 3—figure supplement 1H). These results indicate that SP1 is directly responsible for YEATS2 upregulation in HNC.

YEATS2 and GCDH regulate histone crotonylation in HNC

After establishing the role of YEATS2 as one of the important epigenetic factors involved in maintaining EMT-phenotype in HNC cells, we wanted to delve into the mechanism of YEATS2-mediated regulation of gene expression. Being a histone reader with a varying magnitude of affinity for multiple histone modifications, the function of YEATS2 could be potentially influenced by the metabolic state of the cell since metabolic intermediates can act as substrates for histone modifying proteins. Therefore, in order to explore the influence of cellular metabolism on the dynamics of histone modifications maintained by YEATS2, we wanted to look into the metabolic enzymes that are co-expressed with YEATS2 in HNC. For this, we performed gene set enrichment analysis (GSEA) with a comprehensive list of essential metabolic pathways ¹⁸[↗](#) and found out that KEGG LYSINE DEGRADATION was the top-most pathway associated with high YEATS2 expression in TCGA samples (**Figure 4A** [↗](#), Supplementary Table 3). The saccharopine pathway (canonical pathway for lysine degradation) results in conversion of lysine to 2 molecules of acetyl-CoA, with several intermediates including crotonyl-CoA (**Figure 4B** [↗](#)) ¹⁹[↗](#). Since it is known that YEATS2 has a higher affinity for binding crotonyl groups on histone (specifically on H3K27 residue) than other histone substrates ⁹[↗](#), we hypothesized that the crotonyl-CoA producing enzyme glutaryl CoA dehydrogenase (GCDH), which is a part of lysine degradation pathway, would have increased expression in HNC. We found that *GCDH* was upregulated and had an inverse pattern of expression with the downstream enzyme *ECHS1* in TCGA cancer vs. normal tissue dataset (Figure 4—figure supplement 1A-B). Further, we found that the mRNA expression of *GCDH* was significantly upregulated in head and neck cancer samples relative to their matched normal tissues (**Figure 4C** [↗](#)). Moreover, correlation analysis performed between the expression levels of *GCDH* and YEATS2 in TCGA data showed significant Spearman correlation coefficient value of 0.32 (Figure 4—figure supplement 1C). We then proceeded to check the levels of histone crotonylation (H3K27cr) in a set of nuclear protein lysates obtained from HNC tissues and found that the abundance of this histone mark was higher in tumor as compared to normal (**Figure 4D** [↗](#), Figure 4—figure supplement 1D). Additionally, we performed IHC to evaluate the levels of *ECHS1*, *GCDH*, YEATS2, H3K27cr, and Twist1 in HNC tumor and their matched normal tissue sections. *ECHS1* expression was notably reduced in tumor tissues, whereas *GCDH*, YEATS2, H3K27cr, and Twist1 exhibited increased staining intensity in tumor sections compared to the adjacent normal tissues (**Figure 4E** [↗](#), Figure 4—figure supplement 1E). These findings indicate that the presence of YEATS2 and *GCDH* is possibly leading to an increase in the EMT phenotype (indicated by higher Twist1 expression) in cancer cells by promoting histone crotonylation. Furthermore, our observations suggest that an elevated *GCDH*/*ECHS1* ratio in HNC tissues may be associated with increased levels of histone crotonylation. Since mitochondrial fraction of *GCDH* cannot influence histone crotonylation directly, we checked its subcellular localization in HNC tissues. Using immunohistochemistry, we detected the presence of nuclear *GCDH* in HNC tissues and observed that the regions that stained strongly for nuclear *GCDH*, also showed abundance in H3K27cr (Figure 4—figure supplement 1F). Conversely, the weakly stained regions for nuclear *GCDH* were less abundant in H3K27cr levels. This established an association between the presence of H3K27cr marks and the expression of *GCDH* in HNC. Further, on checking the levels of H3K27cr on YEATS2- and *GCDH*-knockdown using immunoblotting and immunofluorescence, we found that the cells silenced for either of the two genes caused a decrease in H3K27cr levels (**Figure 4F-I** [↗](#), Figure 4—figure supplement 1G-H). On the other hand, an increase in H3K27cr signal was observed when YEATS2 was overexpressed ectopically in BICR10 cells (Figure 4—figure supplement 1I). These findings indicate a coordinated regulation of histone crotonylation in HNC cells by the epigenetic factor YEATS2 and the metabolic enzyme *GCDH*.

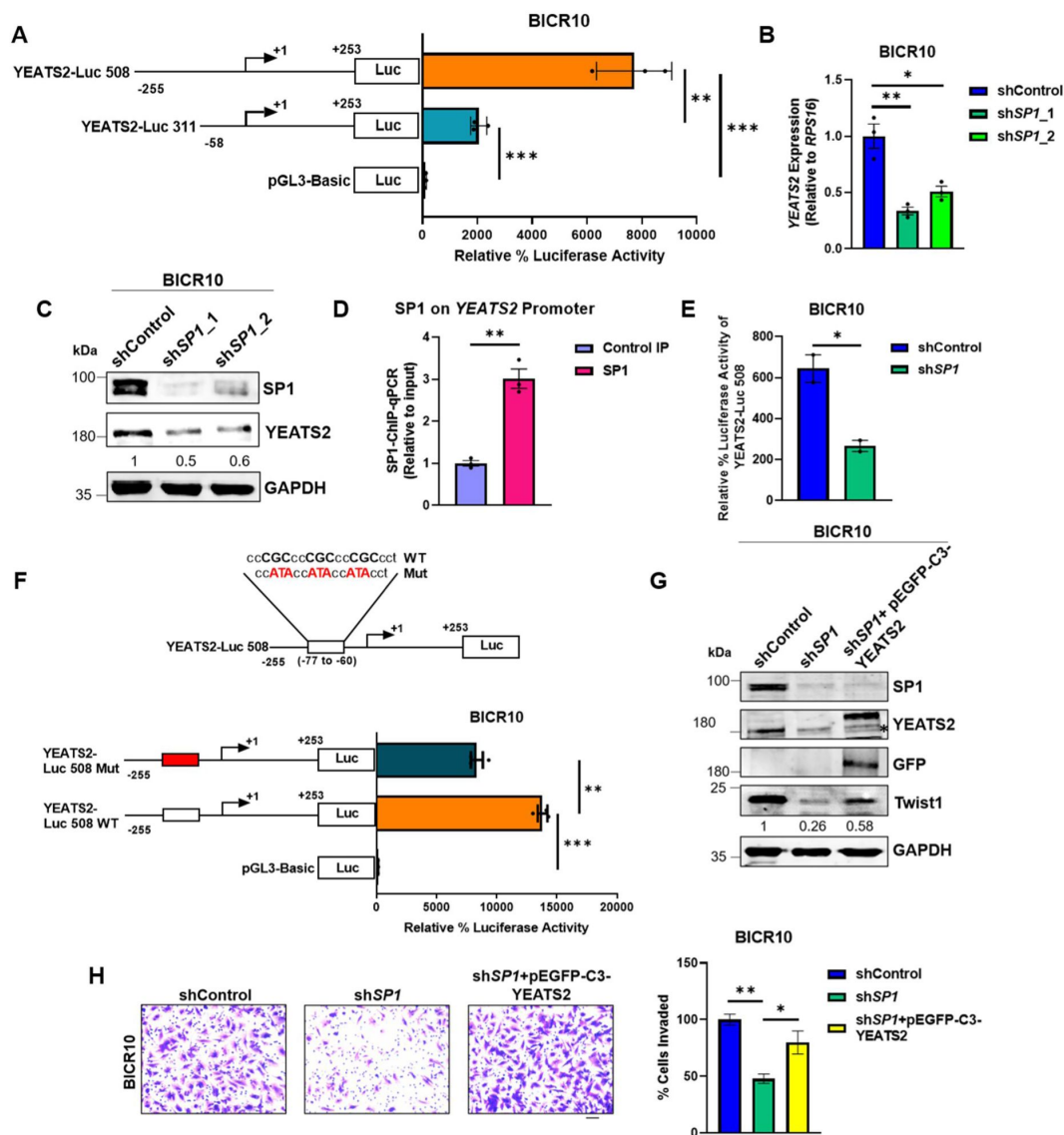


Figure 3.

Regulation of EMT by YEATS2 is SP1-dependent in HNC

(A) Luciferase assay results showing the difference in relative luciferase activity of the two YEATS2 promoter deletion constructs in BICR10. (B) Plot showing decrease in mRNA expression of YEATS2 on SP1-knockdown in BICR10 cells. (C) Immunoblot showing the reduced expression level of YEATS2 upon SP1-knockdown in BICR10. (D) Plot depicting SP1 binding on YEATS2 promoter in SP1-ChIP-qPCR assay in BICR10. (E) Plot showing difference in relative luciferase activity of YEATS2 Luc-508 in shControl vs. shSP1 BICR10 cells. (F) Schematic showing mutation of SP1-binding site sequence in YEATS2 Luc-508 construct (above), and relative luciferase activity of wild-type (WT) vs. mutant (Mut) YEATS2 Luc-508 in BICR10 (below). (G) Immunoblot depicting the decreased Twist1 levels on SP1 knockdown and its subsequent rescue of expression upon YEATS2 overexpression in BICR10 (* indicates endogenous YEATS2 band). (H) Invasion assay images (with quantification on right) showing decrease and rescue of the percentage of invaded cells in shSP1 BICR10 cells, and shSP1 cells with YEATS2 overexpression, respectively. Scale bar, 200 μ m. Error bars, mean $\pm$ SEM; two-tailed t test, * p < 0.05, ** p < 0.01, *** p < 0.001, n = 3 biological replicates.

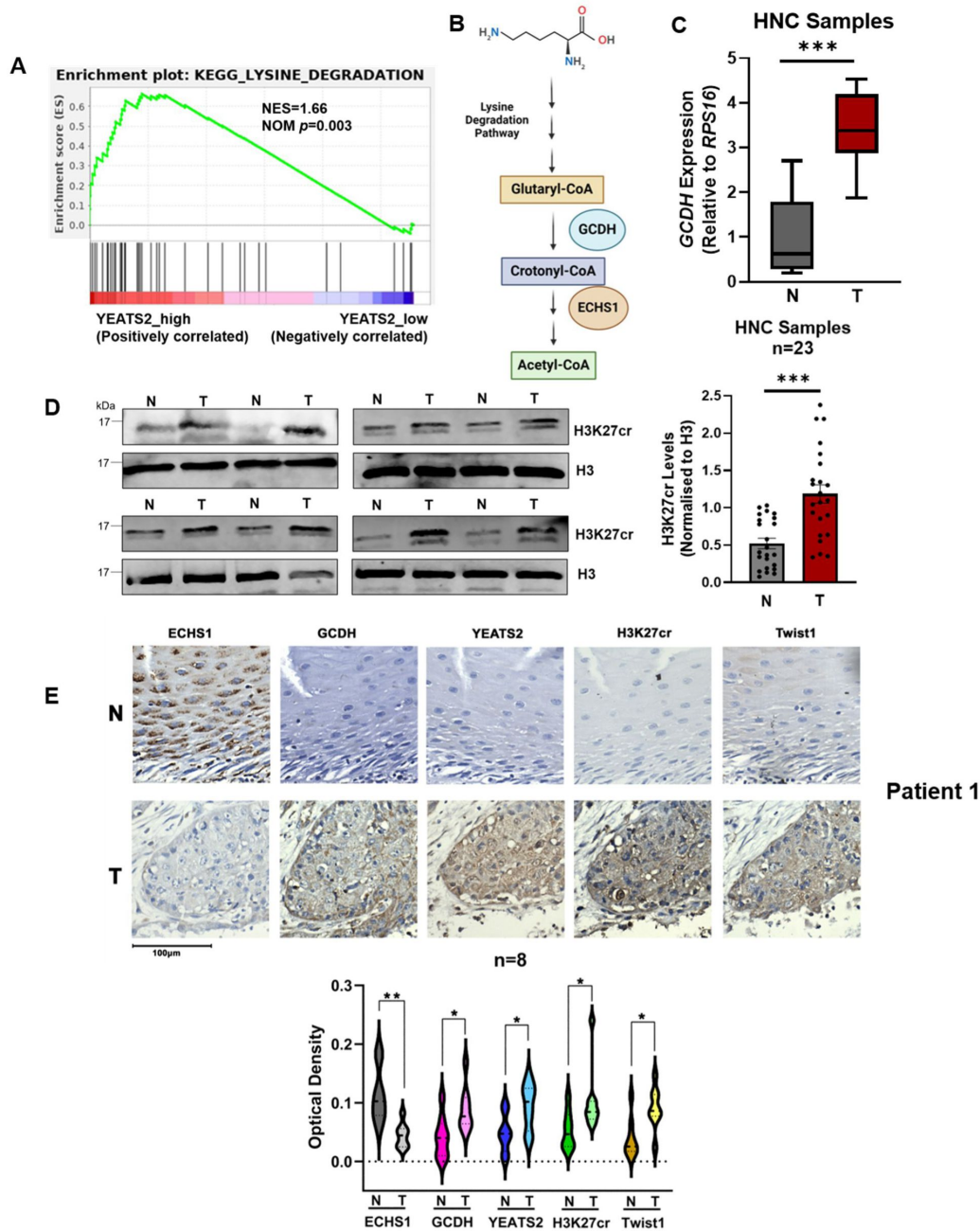


Figure 4.

YEATS2 and GCDH regulate histone crotonylation in HNC.

(A) GSEA plot showing enrichment of KEGG LYSINE_DEGRADATION_PATHWAY in TCGA samples stratified as YEATS2_high as compared to YEATS2_low samples (NES-net enrichment score, NOM p-nominal p-value). (B) Schematic depicting the canonical lysine degradation pathway highlighting the step that leads to crotonyl-CoA production via GCDH. (C) RT-qPCR result showing enhanced mRNA expression of *GCDH* in tumor vs. normal samples (n=8). (D) Immunoblot showing enhanced levels of H3K27cr (quantification on right) in nuclear lysates extracted from HNC tumor vs. normal samples (n=23). (E) Representative IHC images (quantification below) showing the levels of ECHS1, GCDH, YEATS2, H3K27cr and Twist1 in HNC normal vs. tumor tissue samples (n=8) (Scale bar, 100 μm). (F and H) Immunoblot depicting the decrease in H3K27cr levels on (F) YEATS2- and (H) GCDH-knockdown in BICR10 cells. (G and I) Immunofluorescence images depicting the decrease in H3K27cr levels on (G) YEATS2- and (I) GCDH-knockdown in BICR10 cells (Scale bar, 10 μm). Error bars, mean ± SEM; two-tailed t test, **p* < 0.05, ***p* < 0.01, ****p* < 0.001. n = 3 biological replicates.

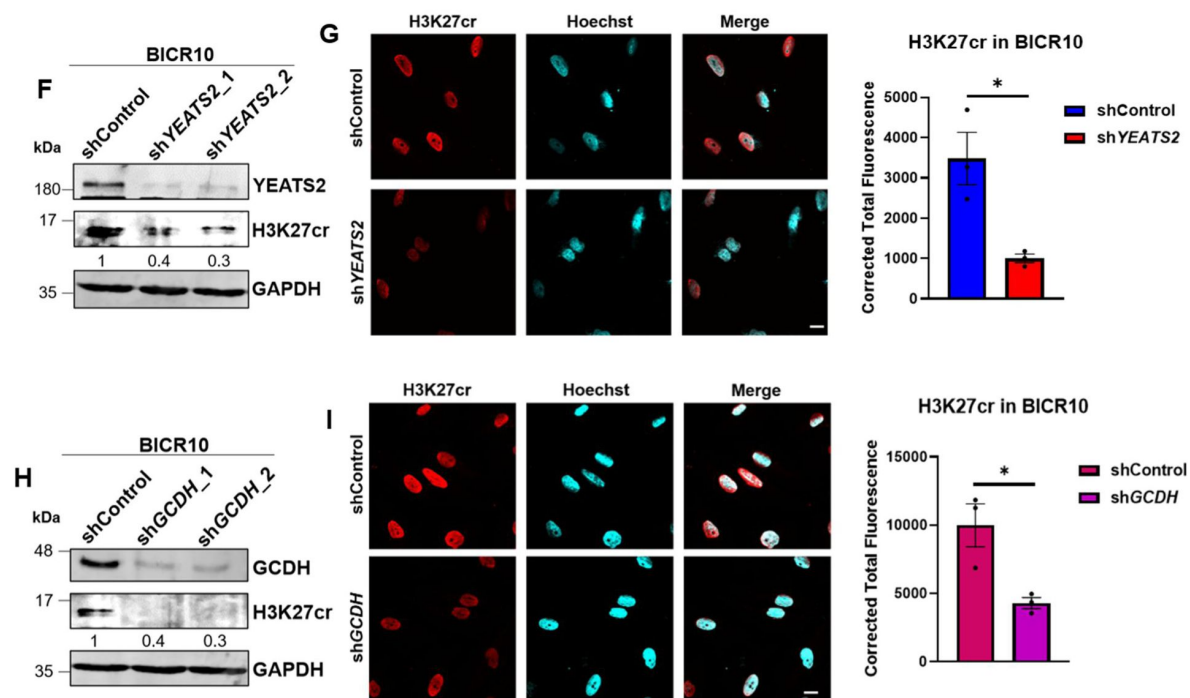


Figure 4. (continued)

YEATS2 regulates expression of EMT-associated gene SPARC

In order to check the effect of YEATS2 on global gene expression, we performed YEATS2 ChIP-seq to obtain genome-wide binding sites of this protein (Figure 5—figure supplement 1A, Supplementary Table 4). We then performed integration of those genes having YEATS2 ChIP-seq peaks on their promoters with the genes that were downregulated in shYEATS2 vs. shControl RNA-seq data, and the genes included in the hallmark EMT gene signature. We found that *SPARC* was the common gene present in all 3 sets (Figure 5A-B, Figure 5—figure supplement 1B). The mRNA expression level of *SPARC* in TCGA dataset was also found to be higher in tumor as compared to normal (Figure 5—figure supplement 1C). Therefore, we chose to further investigate *SPARC* as a direct gene target of YEATS2 in mediating EMT in HNC. Secreted protein acidic and rich in cysteine (SPARC) is a secreted glycoprotein known to be a part of extracellular matrix (ECM). It interacts with various components of ECM and is known to play a role in processes like tissue remodeling and angiogenesis^{20,21}. By performing YEATS2 ChIP-qPCR in YEATS2-downregulated cells, we found that the binding of YEATS2 on *SPARC* promoter decreased on its knockdown in BICR10 as well as SCC9 (Figure 5C, Figure 5—figure supplement 1D). We then checked the expression of *SPARC* at the RNA level on YEATS2-downregulation. *SPARC* expression was decreased in YEATS2-knockdown conditions in both the cell lines (Figure 5D, Figure 5—figure supplement 1E). Since SPARC is a secretory protein, we investigated its level in conditioned media obtained from shControl and shYEATS2 BICR10 cells. We found a decrease in the secreted pool of SPARC on YEATS2-knockdown (Figure 5E). We also observed a decrease in the cellular levels of SPARC protein in SCC9 (Figure 5—figure supplement 1F). Since the binding of a histone reader protein to a gene promoter cannot solely lead to its increased expression, we explored the possibility of the involvement of a writer protein in collaborating with YEATS2 to maintain higher levels of SPARC in HNC. For this, we checked the expression of p300, a known writer of H3K27cr mark²² in humans, in TCGA dataset. We found that p300 (*EP300* gene) had increased expression in tumor vs. normal HNC dataset (Figure 5—figure supplement 1G). On performing p300-ChIP assay, it was found that its binding was reduced in YEATS2-silenced BICR10 cells when compared to the shControl cells, indicating the potential role of p300 in leading to increased expression of SPARC in HNC cells (Figure 5F). To validate our claim, we performed immunoblotting after p300 knockdown and found that the expression of SPARC in supernatant derived from shEP300 BICR10 cells was reduced as compared to shControl cells (Figure 5G). Further, we performed p300 pulldown in a co-immunoprecipitation experiment to check if YEATS2 and p300 function in a protein complex. We discovered the presence of YEATS2 in the p300-immunoprecipitate sample, clearly indicating that YEATS2 and p300 function together as a complex to drive the expression of SPARC (Figure 5H). This set of results indicates a sophisticated epigenetic mechanism promoting increased transcription of the EMT-related gene SPARC in HNC. Finally, we wanted to check the effect of SPARC on invasion. It was observed that the decrease in invasion seen on YEATS2-downregulation was rescued by SPARC overexpression (Figure 5I-J). This indicates that SPARC is a direct target of YEATS2 and acts as a downstream effector of YEATS2 in maintaining invasive phenotype in HNC cells.

Maintenance of H3K27cr marks is dependent on YEATS2-mediated recruitment of p300 on *SPARC* promoter

To investigate the mechanism of YEATS2-mediated gene regulation we checked the status of histone modifications, both acetylation and crotonylation at the H3K27 residue. We performed ChIP-qPCR assay to check the abundance of both the marks at the *SPARC* promoter and found that only H3K27cr mark showed significant downregulation on YEATS2 knockdown (Figure 6A-B, Figure 6—figure supplement 1A-B). This indicated that SPARC expression is potentially controlled by a non-acetyl histone mark i.e., crotonylation. Further, to validate the role of YEATS2 in the transcription of *SPARC* gene, we performed RNA Pol II ChIP-qPCR assay and observed a decrease

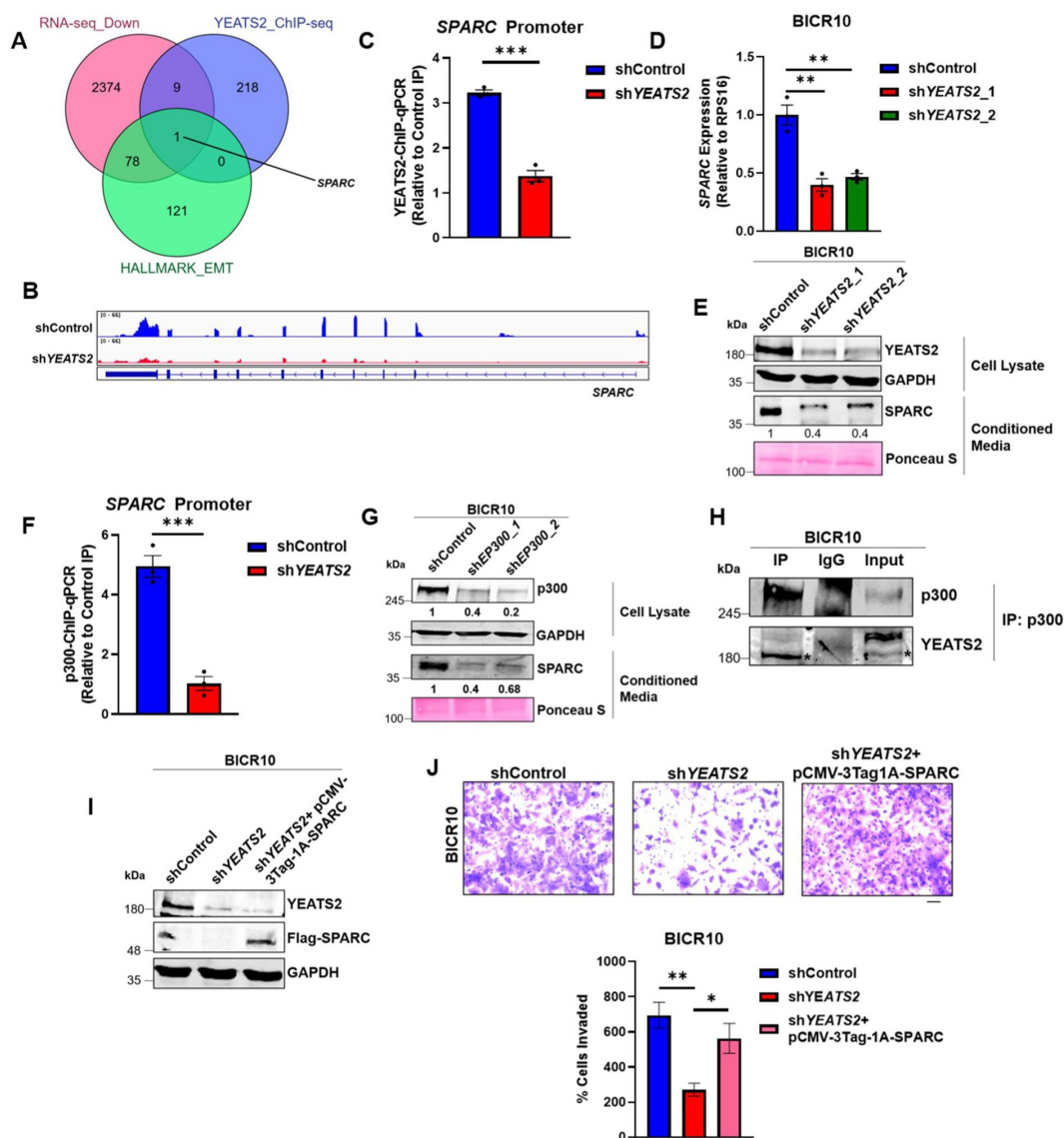


Figure 5.

YEATS2 regulates expression of EMT-related SPARC in HNC.

(A) Venn diagram showing *SPARC* gene obtained after integration of RNA-seq data (genes downregulated in shControl vs. shYEATS2), YEATS2 ChIP-seq data and, hallmark EPITHELIAL_MESENCHYMAL_TRANSITION gene signature. (B) Integrative genome viewer (IGV) plot showing decrease in *SPARC* expression in shControl vs. shYEATS2 RNA-seq data. (C) YEATS2-ChIP-qPCR results showing decreased binding of YEATS2 on *SPARC* promoter in shYEATS2 BICR10 cells. (D) RT-qPCR results showing decreased expression of *SPARC* on YEATS2-knockdown in BICR10. (E) Immunoblot showing decreased expression of *SPARC* in conditioned media derived from shControl and shYEATS2 BICR10 cells. (F) p300-ChIP-qPCR results showing decreased binding of p300 on *SPARC* promoter in shYEATS2 BICR10 cells. (G) Immunoblot showing decreased *SPARC* levels on p300 knockdown in conditioned media from shControl and shEP300 BICR10 cells. (H) Immunoblot showing co-immunoprecipitation of YEATS2 by p300 in YEATS2-overexpressed BICR10 cells (endogenous YEATS2 band is highlighted by *). (I-J) Immunoblot depicting the overexpression of Flag-tagged *SPARC* in shYEATS2 BICR10 cells (I), and Invasion assay images (J) (with quantification below) showing decrease and rescue of the percentage of invaded cells in shControl vs. shYEATS2 BICR10 cells, and shYEATS2 cells with YEATS2 overexpression, respectively. Scale bar, 200 μ m. ** p < 0.01, *** p < 0.001, n = 3 biological replicates.

in Pol II occupancy on *SPARC* promoter after YEATS2 knockdown (**Figure 6C**). Since the substrate used in addition of crotonylation marks on histones is crotonyl-CoA, we supplemented the cells with its precursor molecule sodium crotonate²³, and saw that there was an increase in H3K27cr levels on *SPARC* promoter (**Figure 6D**, Figure 6—figure supplement 1C). We also observed an increase in the expression of *SPARC* at mRNA and protein levels in both the cell lines (**Figure 6E-F**, Figure 6—figure supplement 1D-E). We then checked the effect of crotonate on cancer cell invasiveness and observed that cells treated with crotonate invaded in greater numbers in Matrigel than untreated cells in both the HNC cell lines (**Figure 6G**, Figure 6—figure supplement 1F). Finally, we checked whether crotonate treatment can rescue the effect of YEATS2 downregulation on *SPARC* expression. We did not observe significant rescue either in H3K27cr levels on *SPARC* or in *SPARC* protein expression, when YEATS2-downregulated cells treated with crotonate (**Figure 6H-I**). With these observations, we can conclude that H3K27cr-mediated increase in *SPARC* expression is subject to the availability of YEATS2 in HNC cells.

Since we have previously established the role of p300 in the transcriptional regulation of *SPARC*, we now wanted to investigate the exact mechanism involved behind this regulation. For this, we performed p300 knockdown followed by H3K27cr-ChIP-qPCR and found a significant decrease in H3K27cr levels on *SPARC* promoter in shEP300 BICR10 cells (**Figure 6J**). This indicates that p300 regulates *SPARC* expression by functioning as a crotonyltransferase. Further, we wanted to check whether the binding and subsequent activity of p300 is dependent on the presence of YEATS2 on *SPARC* promoter. As shown in **Figure 3**, YEATS2 expression in HNC cells is dependent on SP1. To examine the dependency of p300 activity on YEATS2, we chose to knockdown SP1 followed by YEATS2 overexpression. As expected, on performing p300-ChIP-qPCR we saw a decrease in p300 occupancy on *SPARC* in SP1 knockdown cells, which was rescued by overexpressing YEATS2 in shSP1 cells (**Figure 6K**). A decrease in SP1 levels would have led to abrogation of YEATS2, which in turn reduced p300 recruitment at the *SPARC* promoter. We observed the same pattern in the H3K27cr levels—decreased enrichment of H3K27cr in shSP1 cells and its subsequent rescue on YEATS2 overexpression (**Figure 6L**). In conclusion, the expression of EMT-related gene *SPARC* is regulated by promoter H3K27cr levels added by crotonyltransferase p300, whose recruitment is dependent on the histone reader YEATS2.

GCDH expression is SP1-dependent and it regulates H3K27cr-mediated *SPARC* expression with YEATS2 synergistically

As previously shown, both YEATS2 and GCDH regulate the levels of histone crotonylation in HNC. So, we hypothesized that the presence of GCDH could be essential for production of crotonyl-CoA, which could be added to histones by histone modification machinery consisting of YEATS2 and p300. On performing knockdown of GCDH, we observed a decrease in *SPARC* at protein and mRNA levels (**Figure 7A-B**, Figure 7—figure supplement 1A). Also, on performing H3K27cr-ChIP assay with GCDH-downregulated cells, we found that H3K27cr marks on *SPARC* promoter reduced relative to control cells (**Figure 7C**). We then sought to explore the possibility of localized histone crotonylation performed by a multi-protein complex consisting of YEATS2 and GCDH. For this, we performed co-immunoprecipitation; however, we did not observe an interaction between GCDH and YEATS2 (Figure 7—figure supplement 1B), ruling out the possibility of the localized production of crotonyl-CoA at a specific gene locus²⁴. We then checked whether both the genes are regulated by the same transcription factor i.e., SP1. TCGA gene expression dataset showed a significant correlation between the expression levels of *SP1* and *GCDH*, hinting at the possibility of SP1 transcriptionally regulating *GCDH* in HNC (Figure 7—figure supplement 1C). Thus, we performed SP1-knockdown and found a decrease in *GCDH* levels in both HNC cell lines (**Figure 7D**, Figure 7—figure supplement 1D). On performing SP1-ChIP we found a significant occupancy of SP1 on *GCDH* promoter in BICR10 and SCC9 (**Figure 7E**, Figure 7—figure supplement 1E). These results suggest that SP1 is responsible for high expression of both *GCDH* and YEATS2 in HNC. As we have already established that YEATS2 abrogation leads to decrease in the invasiveness of HNC cells, we now wanted to check the effect of *GCDH* downregulation on HNC cell invasion. We

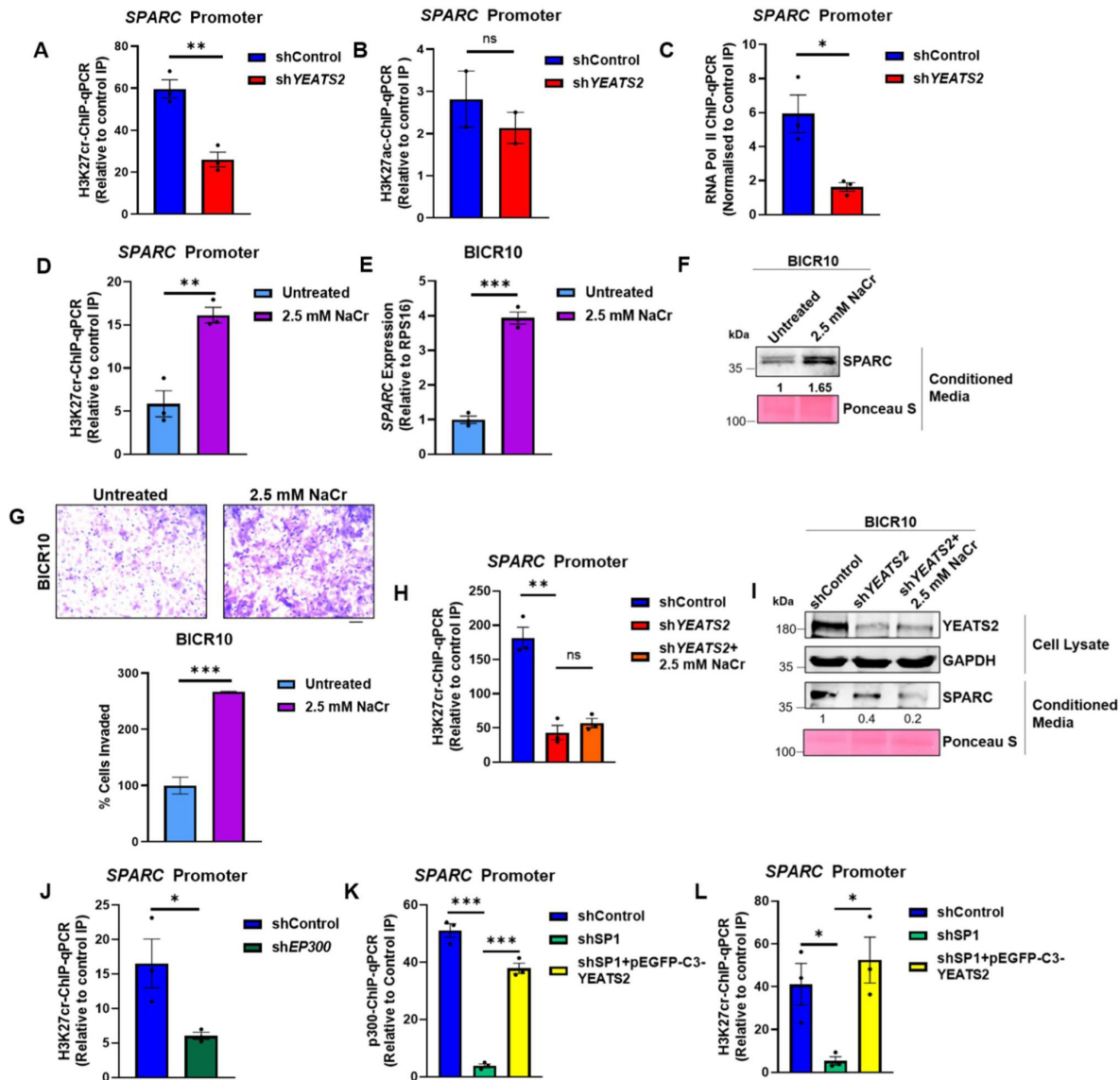


Figure 6.

Maintenance of H3K27cr marks is dependent on YEATS2-mediated recruitment of p300 on SPARC promoter.

(A) H3K27cr-ChIP-qPCR results showing decrease in H3K27cr enrichment on SPARC promoter on YEATS2 knockdown. (B) H3K27ac-ChIP-qPCR results showing non-significant change in H3K27ac enrichment on SPARC promoter on YEATS2 knockdown. (C) RNA Pol II ChIP-qPCR showing decrease in Pol II occupancy on SPARC after YEATS2 downregulation in BICR10. (D) H3K27cr-ChIP-qPCR results showing increase in H3K27cr enrichment on SPARC promoter on treating BICR10 cells with 2.5 mM sodium crotonate (NaCr). (E-F) RT-qPCR (E) and immunoblot (F) showing enhanced SPARC expression in untreated BICR10 cells vs. BICR10 cells treated with 2.5 mM NaCr. (G) Invasion assay images (with quantification below) showing increased invasion on treating BICR10 cells with 2.5 mM NaCr (Scale bar, 200 μ m). (H) H3K27cr-ChIP-qPCR data depicting the lack of significant difference in H3K27cr levels between shYEATS2 vs. shYEATS2+ 2.5 mM NaCr cells. (I) Immunoblot showing inability of NaCr treatment to rescue SPARC expression in YEATS2-knockdown BICR10 cells. (J) H3K27cr ChIP-qPCR showing decrease in H3K27cr levels in BICR10 shControl vs. shEP300 cells. (K-L) p300 (K) and H3K27cr ChIP-qPCR (L) showing decrease and subsequent rescue in p300 binding and H3K27cr enrichment on SP1 knockdown and YEATS2 overexpression, respectively. Error bars, mean $\pm$ SEM; two-tailed t test, ns-non-significant, * p < 0.05, ** p < 0.01, *** p < 0.001.

performed Matrigel invasion assay with shControl and shGCDH BICR10 cells and found a significant decrease in the invasive capacity of shGCDH cells (**Figure 7F**). Conversely, the number of invading cells showed an increase when GCDH was overexpressed in BICR10 cells (Figure 7—figure supplement 1F), indicating that like YEATS2, GCDH is also essential for imparting invasive properties to HNC cells.

The canonical location of GCDH in lysine degradation pathway is mitochondria. However, having detected the presence of nuclear pool of GCDH in HNC tissues (Figure 4—figure supplement 1F), we wanted to investigate the role of YEATS2 in affecting the subcellular localization of GCDH in HNC cells. We detected the presence of GCDH in nucleus in BICR10 using IF, and observed that the levels of nuclear GCDH diminished in shYEATS2 cells as compared to shControl cells (**Figure 7G**). This indicates that YEATS2 might be involved in regulating the transport of GCDH inside nucleus indirectly by an unknown mechanism. In order to further investigate this epigenome-metabolism interplay, we checked the role of YEATS2 and GCDH in regulating SPARC expression in the absence of SP1. The dual overexpression of YEATS2 and GCDH in SP1-silenced cells rescued the SPARC levels to a greater degree; whereas, YEATS2 and GCDH when overexpressed alone, could only partially rescue SPARC expression (**Figure 7H**). Hereby, we establish that one of the axes involved in regulation of EMT in HNC is dependent on the synergistic maintenance of H3K27cr marks on *SPARC* gene by YEATS2 and GCDH.

YEATS2 regulates gene expression by maintaining H3K27cr levels globally

We have previously shown that the expression of a large number of genes are affected by YEATS2 downregulation (**Figure 1H**). For establishing a direct relationship between YEATS2-mediated histone crotonylation and gene expression, we performed ChIP-sequencing to check global H3K27cr levels in shControl and shYEATS2 BICR10 cells. We found that downregulation of YEATS2 led to a significant change in H3K27cr enrichment at 1,583 locations in the genome, out of which 1,533 locations showed significant decrease in H3K27cr enrichment (**Figure 8A**). Further, 1,166 differentially lost sites were found to be present at the promoter regions of genes. As expected, overrepresentation analysis using these set of genes revealed EPITHELIAL MESENCHYMAL TRANSITION as one of the enriched hallmark gene sets (**Figure 8B**). We then overlapped the genes having reduced H3K27cr levels with the genes that showed downregulation in shControl vs. shYEATS2 RNA-seq. A total of 147 genes were found to be common in both the sets (**Figure 8C**, Supplementary Table 5). Among these 147 genes, some examples of EMT-associated genes were *ERG*²⁵ and *CREB3L2*²⁶ (**Figure 8D-E**). Overall, our results suggest that YEATS2 regulates histone crotonylation throughout the genome resulting in the transcriptional activation of a large number of genes, many of which are related to the process of EMT.

Discussion

In our study, we identified YEATS2 as an important epigenetic regulator that plays an oncogenic role in head and neck tumor progression. We have shown that the histone reader regulates expression of important EMT-related genes. The mechanism employed by YEATS2 for gene regulation was also investigated. The YEATS domain confers unique properties to this epigenetic factor, making it suitable for binding with modifications other than acetylation²⁷. As stated earlier, YEATS2 has a binding affinity for a variety of chemical groups. Among all histone modifications screened, YEATS2 had the highest affinity for histone crotonylation, specifically at the histone H3 lysine 27 position⁹. There have been conflicting reports about the role of crotonylation at the H3K27 position. YEATS domain-containing GAS41 was found to recruit SIN3A-HDAC1 complex in a H3K27cr-dependent manner at the promoters of tumor suppressor genes, thereby leading to their repression in colorectal cancer²⁸. Conversely, another study in colorectal cancer showed that high H3K27cr at the promoter of *ETS1* gene led to

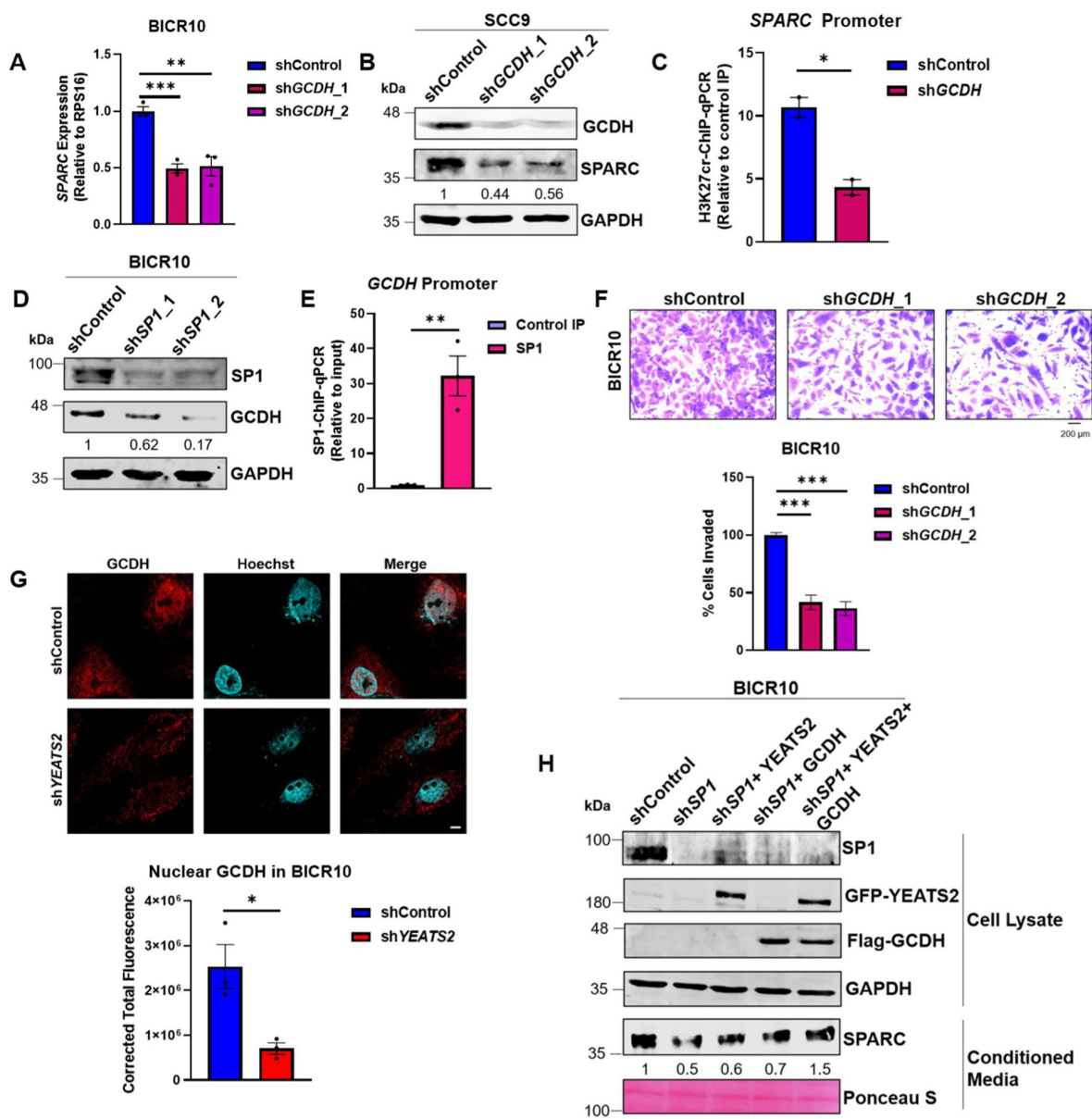


Figure 7.

GCDH expression is SP1-dependent and regulates H3K27cr-mediated SPARC expression with YEATS2 synergistically.

(A) RT-qPCR results showing decrease in *SPARC* expression on GCDH-knockdown in BICR10 cells. (B) Immunoblot showing the reduced expression of *SPARC* upon GCDH-knockdown in SCC9 cells. (C) Decrease in H3K27cr levels on *SPARC* promoter in shGCDH BICR10 cells. (D) Immunoblot showing the reduced expression of GCDH on SP1-knockdown in BICR10 cells. (E) Plot showing SP1 binding on *GCDH* promoter in SP1-ChIP assay in BICR10. (F) Invasion assay images (quantification shown below) showing decrease in invasion of BICR10 cells on GCDH knockdown (Scale bar, 200 μ m). (G) IF images (quantification shown below) showing reduced nuclear localization of GCDH in shYEATS2 BICR10 cells. Scale bar, 5 μ m. (H) Immunoblot depicting decreased *SPARC* expression on SP1 knockdown and its subsequent rescue upon dual YEATS2 and GCDH overexpression in shSP1 BICR10 cells. Error bars, mean $\pm$ SEM; two-tailed t test, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $n = 3$ biological replicates.

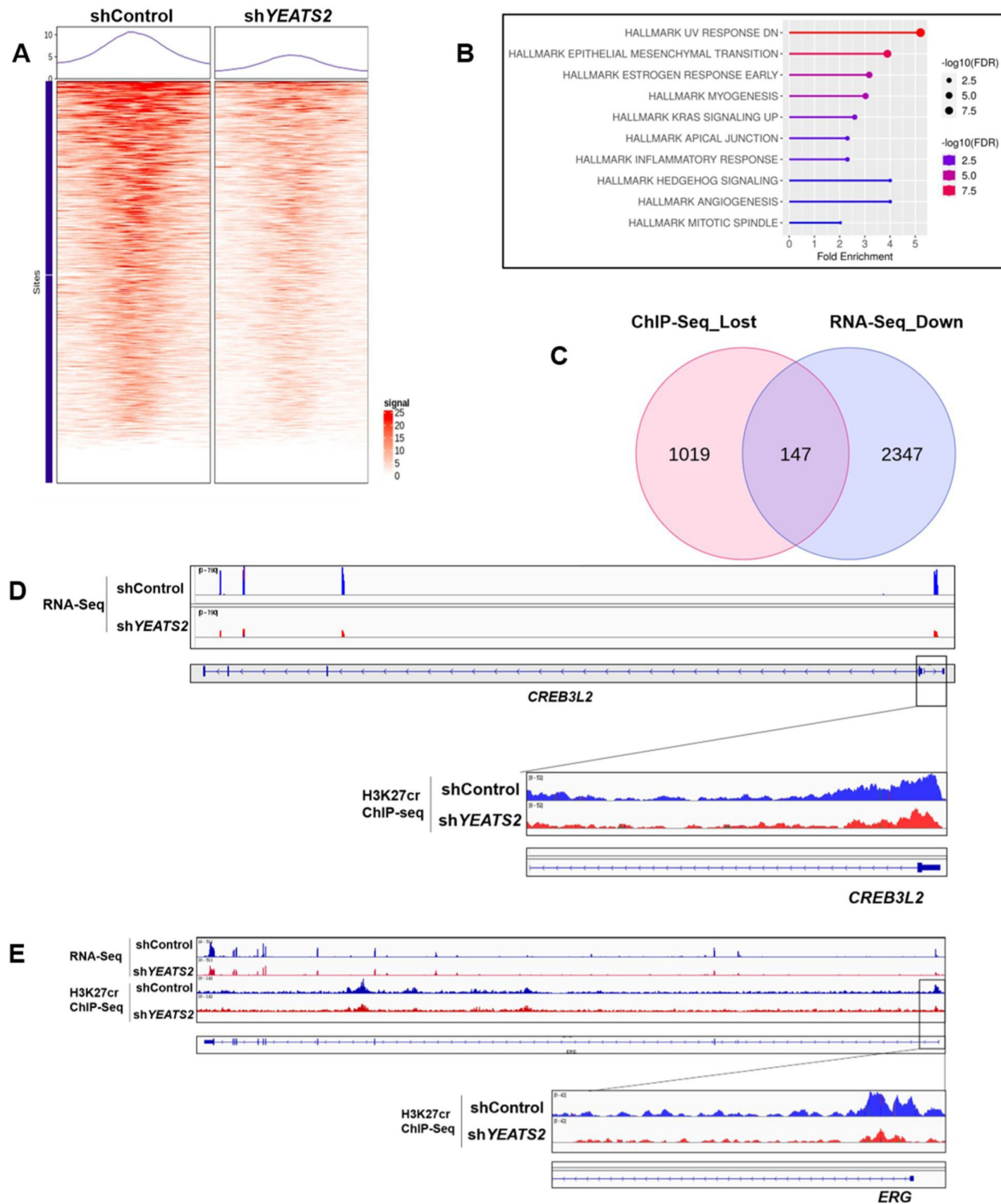


Figure 8.

YEATS2 regulates gene expression by maintaining H3K27cr levels globally.

(A) Differential ChIP-seq profile of H3K27cr in shControl vs. shYEATS2. (B) Overrepresentation analysis showing pathways enriched among genes with decreased H3K27cr enrichment in H3K27cr ChIP-seq data. (C) Overlap of genes with reduced H3K27cr on their promoter and downregulated genes in RNA-seq. (D-E) IGV plot showing representative examples of genes (CREB3L2 and ERG) common in (C).

an increase in its expression and promoted metastasis ¹⁰. In our findings, we have corroborated the latter study by showing the association of H3K27cr with the activation of an EMT-associated gene.

With the discovery of a plethora of novel histone modifications, researchers have been facing the challenge to understand the role of these marks in influencing gene expression. A group of such histone marks, known as the non-acetyl acylation marks, consist of a wide variety of chemical moieties which differ in structure and chemical properties ^{29,30}. Due to the differences amongst these histone marks, it is obvious to assume that the proteins interacting with such histone modifications would not be the same as those interacting with histone acetylation or methylation. YEATS domain is one of the domains that has affinity for non-acetyl histone acylation marks. There are 4 members in the family of YEATS-domain containing proteins, AF9, YEATS2, ENL and, GAS41 ²⁷. By using multiple high-throughput next generation sequencing techniques, we have strengthened the evidence for the role of YEATS2 in upregulating the expression of EMT-associated gene targets by maintaining a non-acetyl histone mark on their promoters. Global abrogation of pro-EMT genes was seen in RNA-seq data of YEATS2-silenced cells, which was underlain by a simultaneous decrease seen in H3K27cr levels observed in H3K27cr ChIP-seq. This established a direct relationship between tumorigenic gene expression and histone crotonylation in cancer. Furthermore, in order to thoroughly investigate this link, we have integrated different sets of NGS data and found that invasion-associated gene *SPARC* gene is a direct target of H3K27cr-mediated regulation by YEATS2, and is possibly one of the many ways in which the EMT-promoting effect of YEATS2 is relayed downstream. As a part of the tumor microenvironment, *SPARC* mediates ECM-tumor cell communication. Several reports have suggested the role of *SPARC* in contributing towards a pro-invasive phenotype in cells multiple cancer types ^{31,32}. Presence of *SPARC* in tumor stroma was reported to be responsible for recruitment of immunosuppressive myeloid cells in high grade breast carcinomas ³³. In this study, we observed that the *SPARC* was capable of rescuing the decrease in invasive properties of YEATS2-silenced cells, making it the functional downstream effector of YEATS2.

We have discovered direct interaction between YEATS2 and epigenetic writer p300 in our study. Research has demonstrated diversity among histone reader families, each exhibiting preferences for specific histone marks. However, considerable overlap exists among histone marks deposited by the same histone writer. Notably, we found that one of the most promiscuous acyltransferases, p300, can deposit H3K27cr marks at the *SPARC* gene promoter in a YEATS2-dependent manner. Using ChIP assays and rescue experiments we showed that p300 binds to *SPARC* promoter and its binding is reduced in the absence of YEATS2. Recent studies have highlighted the importance of p300-mediated crotonylation at H3K27 and other histone residues in the crucial process of early embryonic development. Depletion of p300 led to reduced histone crotonylation resulting in dysregulated gene expression in developing embryos causing them to be arrested at the 8-cell stage ³⁴. Our study adds to the growing body of research underscoring the importance of YEATS2-and p300-mediated histone crotonylation in both physiological and pathological processes.

Our study demonstrates the cooperation between an epigenetic factor and a metabolic enzyme working together to mediate EMT-promoting gene expression. This crosstalk has not been reported in head and neck cancer previously. Since epigenetic changes and aberrant metabolism both are major hallmarks of cancer ^{8,35}, it is important to investigate the metabolism-epigenetics axis in order to yield new therapeutic targets. Increasing evidences suggest that the state of chromatin in a cell is a reflection of the metabolic enzymes functioning inside the nucleus. Through gene set enrichment analysis, we found that there was a significant correlation between YEATS2 and the lysine degradation pathway in HNC. This led to the observation that one of the enzymes crucial for the production of crotonyl-CoA, GCDH is overexpressed in HNC. Previous reports have suggested that metabolic enzymes could affect epigenomic changes by functioning closely with epigenetic factors. Studies showing the presence of multi-subunit complexes consisting of a metabolite-producing enzyme, DNA-binding protein and, a histone writer are a

testament to the fact that metabolism-epigenome collaboration is more direct than previously thought³⁶. Although we did not observe direct association of GCDH and YEATS2, we did establish that the nuclear presence of GCDH causes an increase in histone crotonylation in HNC. As per the evidence provided by us, it could be proposed that the surge in GCDH in YEATS2-high cancer cells affects the chromatin by increasing the pool of crotonyl-CoA inside nucleus, which acts as a substrate for histone modifying enzymes. Since YEATS2 works by recruiting p300 via recognition of histone marks, maintenance of crotonylation at promoter regions could be dependent on sites with a constitutive basal level of H3K27cr or other marks, as a histone reader cannot recognize a genomic locus devoid of pre-existing histone marks. Such constitutive sites would be used by YEATS2 as an anchor to recruit p300 and subsequently induce histone crotonylation on nearby “inducible” sites. Further investigation is needed to precisely locate and distinguish between constitutive and inducible sites of histone crotonylation. Moreover, we have shown that SP1 transcriptionally regulates both YEATS2 and GCDH. This hints towards a possibility of a direct axis initiated by an unknown upstream factor working in order to modify the chromatin in such a manner that it benefits the cancer cell by causing an increase in the tumorigenic properties of the cell.

In conclusion, we have presented a sophisticated interplay between YEATS2 and GCDH, where the upregulation of both factors contributes towards the enhancement of EMT in head and neck cancer by changes in the global EMT-favoring gene expression, mediated through promoter histone crotonylation (**Figure 9**). Our findings provide new insights into the metabolic-epigenetic crosstalk driving EMT and identify the YEATS2–GCDH–SPARC axis as a potential therapeutic target in HNC.

Methods

Cell culture

Human head and neck cancer cell lines BICR10 and SCC9 were obtained from European Collection of Authenticated Cell Cultures (ECACC). HEK293T cell line was obtained from American Type Cell Culture (ATCC). Cells were cultured in media recommended by ECACC and ATCC, supplemented with 10% FBS (Sigma, F7524), 100 units/mL of penicillin and streptomycin (Invitrogen, 15140122) and 4 mmol/L L-glutamine (Sigma, G7513). All cell lines were cultured in a humidified atmosphere at 37°C and 5% CO₂. Transfection of overexpression- and promoter-deletion-constructs was performed using polyethylenimine (PEI) (Polysciences, 23966). For crotonate supplementation, cells were treated with 2.5 mM sodium crotonate for 36 h before harvesting.

Head and neck cancer sample collection

The study was approved by the Institute Ethics Committee of the Indian Institute of Science Education and Research Bhopal, India. Head and neck cancer samples were collected after obtaining informed consent from patients undergoing surgery at Bansal Hospital, Bhopal, India. Details of patients are given in Supplementary Table 6.

Bioinformatics analyses

In order to study cancer associated epigenetic factors we analyzed The Cancer Genome Atlas (TCGA) mRNA expression dataset using TCGAbiolinks package³⁷ in R to obtain epigenetics-related genes that are significantly upregulated in head and neck cancer as compared to normal tissues. Survival analysis was then performed using an online tool OncoLnc (<http://www.oncolnc.org>). Results of this analysis are given in Supplementary Table 1. To obtain the expression of individual genes in TCGA, online platforms like Gepia2³⁸ and UALCAN³⁹. Correlation analysis was done using Gepia2; EMT hallmark signature (EPITHELIAL_MESENCHYMAL_TRANSITION) used was downloaded from MSigDb database⁴⁰. GSEA was performed as described previously

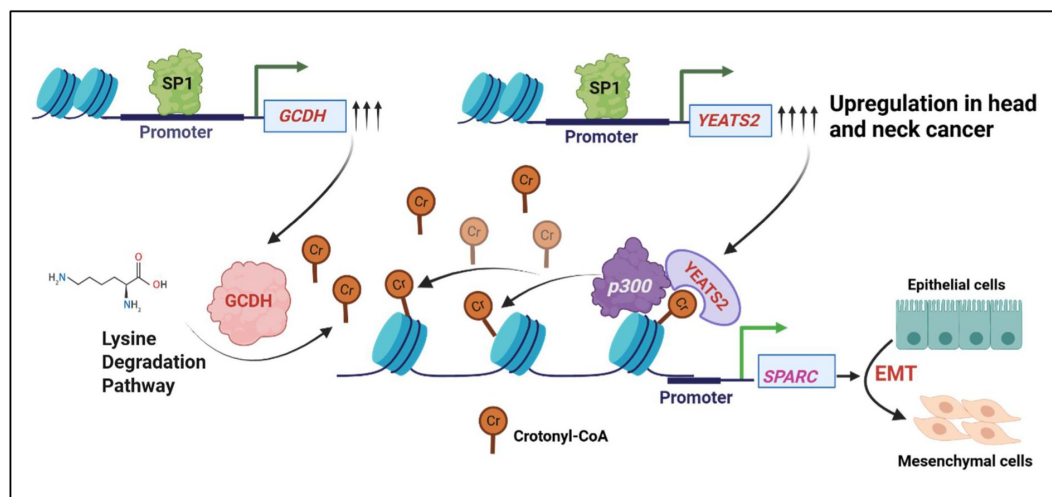


Figure 9.

Schematic showing YEATS2- and GCDH-mediated regulation of EMT through H3K27cr-dependent SPARC upregulation in head and neck cancer.

⁴¹ Stratification of TCGA gene expression data on the basis of YEATS2 expression was done and the enrichment of various metabolic pathways¹⁸ in the YEATS2_high vs. YEATS2_low group was performed using GSEA software⁴². Results of this analysis are provided in Supplementary Table 3.

RNA interference

SCC9 and BICR10 cells (2×10^5) were seeded in a 6-well plate. After 24 h of seeding, cells were infected with lentivirus containing small hairpin RNA (shRNA) (Sigma, Mission Human Genome shRNA Library) against *YEATS2*, *SP1*, *EP300* and *GCDH* or shControl plasmid with 8 µg/mL polybrene (Sigma, H9268) containing media. Selection was then done using puromycin at 1 µg/mL (Sigma, P9620) for 3 days before harvesting. Sequences of shRNA are provided in Supplementary Table 8.

Molecular cloning

The *YEATS2* gene was cloned in pEGFP-C3, whereas *SPARC*, *SP1*, and *GCDH* were cloned in pCMV-3Tag-1A overexpression plasmid (amplified using Phusion Polymerase, NEB, M0530S). *YEATS2* was cloned between SalI and SmaI sites, *SPARC* between BamHI and HindIII sites, *SP1* between EcoRI and SalI sites, and *GCDH* between HindIII and SalI sites. The construct was then confirmed using colony PCR and Sanger sequencing. Promoter deletion constructs of *YEATS2* were cloned in pGL3-Basic vector between KpnI and HindIII RE sites. cDNA or genomic DNA from BICR10 cells was used as template in all PCR reactions.

Site directed mutagenesis

The site-directed mutant construct of the *YEATS2* promoter-deletion construct was prepared by using a strategy previously described⁴³. Oligonucleotides with mutations in the SP1-binding sites present from -71 bp to -60 bp upstream of the *YEATS2* TSS (ccCGCccCGCccCGCct to ccATAccATAccATAccct) were used. The wild-type *YEATS2* Luc-508 promoter-deletion construct was used as a template. The primers used are given in Supplementary Table 7. The mutation was confirmed by DNA Sanger sequencing.

Luciferase assay

BICR10 and SCC9 cells (2.5×10^5) were seeded in 6 well plate. After 24 h, cells were co-transfected with either *YEATS2*-promoter deletion wild-type constructs or SP1-site mutated constructs and pRL-TK Renilla (Promega, E2231) luciferase plasmid. After 24 h of transfection cells were lysed using passive lysis buffer (1% Triton X-100, 25mM Tricine pH 7.8, 15mM potassium phosphate pH 7.8, 15mM magnesium sulphate, 4mM EGTA, 1mM DTT). The Firefly/ Renilla luciferase activity was measured using a luminescence detecting plate reader (CYTATION 5, BioTeK).

Transwell invasion assay

BICR10 and SCC9 (2×10^4) cells resuspended in FBS-free media were added to the upper chamber of transwell setup over a Matrigel layer (Corning, 356230) and incubated for 24 h in a cell culture incubator. Lower chamber of the setup contained media supplemented with FBS. The non-migrated cells in the upper layer of Matrigel were removed, and cells migrated to the lower chamber of transwell were fixed in 4% formaldehyde and then stained with crystal violet (0.05% in 10% methanol in $1 \times$ PBS). Five random fields were counted using an inverted microscope (Olympus CKX41).

Wound healing assay

BICR10 and SCC9 (1.2×10^5) cells were seeded in a 12-well plate. 12 h after the cells had attached, a scratch was made in the center of each well and wound healing was monitored for 2 days. Three random regions per well were monitored for migration using EVOS FL Auto 2 Imaging System by

Thermo Scientific.

3D invasion assay

BICR10 and SCC9 (1×10^4) cells were seeded in 96 Well Round (U) Bottom Plate. 48 hours after cell seeding, spheres were transferred to a different flat bottom 96 well plate coated with collagen matrix (Rat Tail Collagen I, Corning 354236) and invasion was checked after 4 days. Images were taken using Olympus CKX41 microscope.

Immunoblotting

Cells were lysed using urea lysis buffer (8M urea, 2M thiourea, 2% CHAPS, 1% DTT) supplemented with $1\times$ protease inhibitor cocktail (0.5 mM Leupeptin, 250 μ M pepstatin, 50 mM EDTA, 1 mM PMSF) at 4°C for 30 min and spun at maximum speed in a 4°C centrifuge for 1 hr. Equal amount of samples were loaded on denaturing polyacrylamide gel and transfer was done to activated PVDF membrane. After transfer, the blots were incubated with recommended dilutions of primary antibodies overnight at 4°C, followed by 40 min incubation with secondary antibody. The blots were scanned using Odyssey Membrane Scanning System. Antibodies used: YEATS2 (Proteintech, 24717-1-AP), SP1 (CST, 9389S), SPARC (CST, 5420S), GCDH (Sigma, HPA043252), N-Cadherin (Abcam, ab19348), Vimentin (Abcam, ab137321), Twist1 (CST, 46702S), GAPDH (CST, 5174S), GFP (Affinity Biosciences, T0006), Flag-tag (Novus, NBP1-06712SS), H3K27cr (PTM Bio, PTM-545RM), Histone H3 (Active Motif, 61475), p300 (CST, 54062S).

Co-immunoprecipitation

Cells were lysed in Co-IP lysis buffer (50 mM Tris HCl pH 7.5, 150 mM NaCl, 1 mM EDTA, 10% glycerol, 1% NP-40) at 4°C for 1 h. Lysate was spun at maximum speed at 4°C for 1 h in a table-top centrifuge. Co-IP mix for p300 IP and control IP was prepared by taking 1 mg of lysate and diluting it with Co-IP lysis buffer to make a total volume of 500 μ l. Antibody for IP (p300, CST:54062S) and control IP (Normal Rabbit IgG, CST: 2729S) was added to the respective tubes and incubated overnight at 4°C. After incubation, 15 μ l of Protein G Dynabeads (Invitrogen, 10004D) was added to each tube and incubated for 2 h at 4°C, followed by washing using Co-IP lysis buffer. The captured protein complexes were eluted by adding Laemmli buffer to the beads. The tubes containing the beads were then heated at 95°C for 5 min. The eluant was loaded on SDS-PAGE gel and immunoblotting was performed using standard protocol.

Immunofluorescence

Cells were plated at a seeding density of 2×10^5 on a coverslip in a 6-well plate. Next day, cells were washed thrice with ice-cold PBS followed by 4% formaldehyde fixation and permeabilization with 0.1% Triton X-100 for 15 min. Blocking was done for 1 h at RT using 2% BSA solution. Cells were then incubated with primary antibodies overnight at 4°C (H3K27cr at 1:200; GCDH at 1:50). Cells were washed thrice with PBS and incubated with Alexa-Fluor 555 anti-rabbit IgG secondary antibody for 1 h at RT. The cells were then counterstained with Hoechst 33342 (Invitrogen, HI399) and mounted using fluoroshield (Sigma, F6182). Imaging was performed using Olympus FV3000 live cell microscope with a 60 $\times$ and 100 $\times$ oil objective lenses, and image analysis was done using Image J software.

Immunohistochemistry

Formalin-fixed, paraffin-embedded human head and neck cancer tissue sections were obtained from Bansal Hospital, Bhopal, India. Clinical characteristics of patients used in the study are presented in Supplementary Table 6. As described previously⁴⁴, slides were kept at 65°C for 2 h on hot plate, deparaffinized and, rehydrated as described previously. Heat induced antigen retrieval was then performed using 10 mM sodium citrate buffer (pH 6 in microwave for 14 min). 1:10 dilution of 3% hydrogen peroxidase in methanol was used to quench endogenous peroxidase.

Blocking was done with 3% BSA. Primary antibodies against ECHS1 (Sigma, HPA022476; 1:50 dilution), H3K27cr (1:200), GCDH (1:50), YEATS2 (1:50), and Twist1 (1:50) were used. HRP/DAB-chromogenic based Super Sensitive Polymer-HRP Detection System kit (BioGenex, QD430-XAKE) was used as per manufactures' instructions for staining. All slides were counterstained with Harris' hematoxylin (Merck). The images were captured by Thermo Scientific Invitrogen EVOS FL Auto 2 Imaging System. Images were then processed in Adobe Photoshop Version 7.0.

Chromatin immunoprecipitation

ChIP assay was performed as described previously⁴⁵. Briefly, cells were fixed with 1% formaldehyde and quenched by 0.125 M glycine followed by cell lysis. Further, cross-linked chromatin was sonicated to a chromatin fragment length of 100–400 bp approximately and chromatin was immunoprecipitated using YEATS2, H3K27cr, H3K27ac, SP1 and p300 antibody or control (immunoglobulin G) antibody overnight at 4°C followed by addition of Protein G Dynabeads (Invitrogen, 10004D). The obtained complex and input DNA were de-crosslinked in TE buffer with 1% SDS and proteinase K (Invitrogen, 25530049) at 65°C for overnight. Eluted DNA was then purified using PCR purification kit (QIAGEN, 28106). The immunoprecipitated (IP) DNA and 5% input were analyzed by qPCR using locus specific primers (Supplementary Table 7) and SYBR Green Master Mix (Promega, A6002). IP DNA values were normalized to input using the following formula: $2^{-(Ct_{input} - Ct_{IP})}$. Resultant values were subsequently normalized to IgG control IP DNA values. The relative values are represented as mean $\pm$ SEM of triplicates.

RT-qPCR

Total was extracted using TRIzol (Invitrogen, 15596026) according to the manufacturer's instructions. cDNA was synthesized from 3 μ g of total RNA by PrimeScript 1st strand cDNA Synthesis Kit (TaKaRa, 6110A). Amplification reactions were performed on light cycler 480 II (Roche) using Go tag SYBR Green master mix (Promega, 75665). The gene expression was calculated using the following formula: $2^{-(Ct_{control} - Ct_{target})}$ where RPS16 is taken as control. Sequence of each primer used is given in Supplementary Table 7.

RNA-seq

RNA from shControl/ shYEATS2 BICR10 cells was extracted using TRIzol. QIAseq Stranded mRNA Lib Kit (Qiagen, 180441) kit was used to prepare libraries. Paired end sequencing (2 $\times$ 150 bp) of these libraries was performed on Novaseq 6000 platform. Fastq files obtained after sequencing were aligned to GRCh38 human genome using STAR aligner⁴⁶. Read counting for each sample was done using HTseq2⁴⁷. Differential expression analysis was then performed using DESeq2⁴⁸. Genes with significant change (FDR corrected p-value < 0.05) in expression are listed in Supplementary Table 2.

ChIP-seq

Eluted DNA obtained after performing ChIP assay in BICR10 cells using YEATS2 antibody. TruSeq ChIP Library Preparation Kit (IP-202-1012) was used to prepare libraries. Paired end sequencing (2 $\times$ 100 bp) of these libraries was performed on Novaseq 6000 platform. Fastq files obtained after sequencing were aligned to GRCh38 human genome using STAR aligner. Peak calling was performed using MACS2 with '--keep-dup=all' parameter⁴⁹. Genomic distribution of YEATS2 peaks was obtained using ChIPSeeker. For visualizing YEATS2 ChIP-seq signal over SPARC promoter, both replicate BAM files were merged using 'samtools merge'⁵⁰ and then bigWig file was generated using 'bamCoverage' function of deepTools⁵¹. Genes having YEATS2 peak in their promoter region were obtained using Biomart by Ensembl (listed in Supplementary Table 4). For H3K27cr ChIP-seq, ChIP assay was performed using chromatin from shControl and shYEATS2 BICR10 cells. Libraries were made using NEBNext Ultra II DNA library preparation for Illumina Kit (NEBNext, E7645S) and paired end sequencing (2 $\times$ 150 bp) was performed on Novaseq 6000

platform. Fastq files were aligned to hg38 genome using Bowtie2⁵², followed by peak calling with MACS2 and differential binding analysis using DiffBind⁵³. Differential sites with p-value < 0.05 were considered significant.

Statistical analysis

Data are presented as mean $\pm$ SEM unless otherwise stated. At least three independent biological replicates have been performed for each experiment. All statistical analyses were conducted using GraphPad Prism 8 software. Statistical tests used and specific *p* values are indicated in the figure legends.

Data availability

YEATS2 and H3K27cr ChIP-Seq data is deposited at GEO database (GSE275975). RNA sequencing data from shControl and shYEATS2 cells has been deposited at GEO database (GSE275977). This study does not report any original code.

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Additional information

Author contributions

S.S. and D.P. conceived and designed the study. D.P., P.K., R.J. and A.S. performed the experiments and analysed the results. D.P. wrote the manuscript with inputs from all other authors. S.S. supervised the experiments and manuscript preparation. S.S. acquired funding for the study.

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Additional files

Supplementary Figures [↗](#)

Supplementary Tables 1-8 [↗](#)

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Author information

Deepak Pant

Department of Biological Sciences, Indian Institute of Science Education and Research
Bhopal, Bhopal, India
ORCID iD: [0000-0002-7925-9394](#)

Parik Kakani

Department of Biological Sciences, Indian Institute of Science Education and Research
Bhopal, Bhopal, India
ORCID iD: [0000-0001-5142-7774](#)

Rushikesh Joshi

Department of Biological Sciences, Indian Institute of Science Education and Research
Bhopal, Bhopal, India

Abin Sabu

Department of Biological Sciences, Indian Institute of Science Education and Research
Bhopal, Bhopal, India

Shruti Agrawal

Department of Pathology, Bansal Hospital, Bhopal, India

Atul Samaiya

Department of Surgical Oncology, Bansal Hospital, Bhopal, India

Sanjeev Shukla

Department of Biological Sciences, Indian Institute of Science Education and Research
Bhopal, Bhopal, India
ORCID iD: [0000-0003-3361-0588](https://orcid.org/0000-0003-3361-0588)

For correspondence: sanjeevs@iiserb.ac.in

Editors

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Joint Public Review:

This manuscript investigates a mechanism between the histone reader protein YEATS2 and the metabolic enzyme GCDH, particularly in regulating epithelial-to-mesenchymal transition (EMT) in head and neck cancer (HNC).

The authors addressed most of the concerns of the reviewers. They have:

- (1) Increased the patient cohort size from 10 to 23 for evaluating the levels of YEATS2 and H3K27cr.
- (2) Checked the expression of major genes involved in the YEATS2-mediated histone crotonylation axis (YEATS2, GCDH, ECHS1, Twist1, along with H3K27cr levels) in head and neck cancer tissues using immunohistochemistry.
- (3) Analyzed publicly available head and neck cancer patient datasets, which revealed a significant positive correlation between YEATS2 expression and increasing tumor grade.
- (4) Performed GSEA on TCGA HNC patient samples stratified by high versus low YEATS2 expression. This analysis robustly demonstrated a positive enrichment of metastasis-related gene sets in the high YEATS2 expression group, compared to the low YEATS2 group.
- (5) Performed extensive experiments to look into the role of p300 in assisting YEATS2 in regulating promoter histone crotonylation. The p300 was knocked down in BICR10 cells, followed by immunoblotting to assess SPARC protein levels.
- (6) Performed co-immunoprecipitation assays to check for an interaction between endogenous YEATS2 and p300. The results clearly demonstrate the presence of YEATS2 in the p300-immunoprecipitate sample, indicating that YEATS2 and p300 physically interact and likely function together as a complex to drive the expression of target genes like SPARC.
- (7) Performed RNA Polymerase II ChIP-qPCR on the SPARC promoter in YEATS2 knockdown cells.
- (8) To confirm p300's specific role in crotonylation at this locus, they performed H3K27cr ChIP-qPCR after p300 knockdown.

(9) Performed SP1 knockdown (which reduces YEATS2 expression) followed by ectopic YEATS2 overexpression, and then assessed p300 occupancy and H3K27cr levels on the SPARC promoter.

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Author response:

The following is the authors' response to the original reviews.

We sincerely appreciate the editors for overseeing an efficient review process and for upholding the high standards of the journal. We have made extensive revisions to the manuscript after carefully reviewing the reviewers' comments. We have addressed all the comments in our response and have incorporated the changes suggested by the reviewers to the best of our abilities. Notably, we have made the following major changes to the manuscript:

(1) We have increased the patient cohort size from 10 to 23 for evaluating the levels of YEATS2 and H3K27cr.

(2) To further strengthen the clinical relevance of our study, we have checked the expression of major genes involved in the YEATS2-mediated histone crotonylation axis (YEATS2, GCDH, ECHS1, Twist1 along with H3K27cr levels) in head and neck cancer tissues using immunohistochemistry.

(3) We have performed extensive experiments to look into the role of p300 in assisting YEATS2 in regulating promoter histone crotonylation.

The changes made to the manuscript figures have been highlighted in our response. We have also updated the Results section in accordance with the updated figures. Tables 1-4 and Supplementary files 1-3 have been moved to one single Excel workbook named 'Supplementary Tables 1-8'. Additional revisions have been made to improve the overall quality of the manuscript and enhance data visualization. These additional changes are highlighted in the tracked changes version of the manuscript.

Our response to the Public Reviews and 'Recommendations to the Authors' can be found below.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This manuscript investigates a mechanism between the histone reader protein YEATS2 and the metabolic enzyme GCDH, particularly in regulating epithelial-to-mesenchymal transition (EMT) in head and neck cancer (HNC).

Strengths:

Great detailing of the mechanistic aspect of the above axis is the primary strength of the manuscript.

Weaknesses:

Several critical points require clarification, including the rationale behind EMT marker selection, the inclusion of metastasis data, the role of key metabolic enzymes like ECHS1,

and the molecular mechanisms governing p300 and YEATS2 interactions.

We would like to sincerely thank the reviewer for the detailed, in-depth, and positive response. We have implemented constructive revisions to the manuscript to address the reviewer's concerns effectively.

Major Comments:

(1) The title, "Interplay of YEATS2 and GCDH mediates histone crotonylation and drives EMT in head and neck cancer," appears somewhat misleading, as it implies that YEATS2 directly drives histone crotonylation. However, YEATS2 functions as a reader of histone crotonylation rather than a writer or mediator of this modification. It cannot itself mediate the addition of crotonyl groups onto histones. Instead, the enzyme GCDH is the one responsible for generating crotonyl-CoA, which enables histone crotonylation. Therefore, while YEATS2 plays a role in recognizing crotonylation marks and may regulate gene expression through this mechanism, it does not directly catalyze or promote the crotonylation process.

We thank the reviewer for their insightful comment regarding the precision of our title. We agree that the initial wording 'mediates' could imply a direct enzymatic role for YEATS2 in histone crotonylation, which is indeed not the case. As the reviewer correctly points out, YEATS2 functions as a 'reader' of histone crotonylation marks.

However, our research demonstrates that YEATS2 plays a crucial indirect regulatory role in the establishment of these crotonylation marks. Specifically, our data indicates that YEATS2 facilitates the recruitment of the histone crotonyltransferase p300 to specific gene promoters, such as that of SPARC. This recruitment mechanism directly impacts the localized deposition of crotonyl marks on nearby histone residues. Therefore, while YEATS2 does not directly catalyze the addition of crotonyl groups, its presence and interaction with p300 are essential for the regulation and establishment of histone crotonylation at these critical sites.

To accurately reflect this nuanced, yet significant, regulatory mechanism, we have revised the title. We are replacing 'mediates' with 'regulates' to precisely convey that YEATS2 influences the histone crotonylation process, albeit indirectly, through its role in recruiting the enzymatic machinery. The updated title will now read: 'Interplay of YEATS2 and GCDH regulates histone crotonylation and drives EMT in head and neck cancer.' We believe this change maintains the core message of our findings while enhancing the scientific accuracy of the title.

(2) The study suggests a link between YEATS2 and metastasis due to its role in EMT, but the lack of clinical or pre-clinical evidence of metastasis is concerning. Only primary tumor (PT) data is shown, but if the hypothesis is that YEATS2 promotes metastasis via EMT, then evidence from metastatic samples or in vivo models should be included to solidify this claim.

We thank the reviewer for their valuable suggestion regarding the need for clinical or pre-clinical evidence of metastasis. We fully agree that direct evidence linking YEATS2 to metastasis would significantly strengthen our claims, especially given its demonstrated role in EMT.

Our primary objective in this study was to meticulously dissect the molecular mechanisms by which YEATS2 regulates histone crotonylation and drives EMT in head and neck cancer. We have provided comprehensive upstream and downstream molecular insights into this process, culminating in a clear demonstration of YEATS2's functional importance in promoting EMT through multiple in vitro phenotypic assays (e.g., Matrigel invasion, wound

healing, 3D invasion assays). As the reviewer notes, EMT is a widely recognized prerequisite for cancer metastasis[1]. Therefore, establishing YEATS2 as a driver of EMT directly implicates its potential role in metastatic progression.

To further address the reviewer's concern and bridge the gap between EMT and metastasis, we have performed additional analyses that will be incorporated into the revised manuscript:

Clinical Correlation with Tumor Grade: We analyzed publicly available head and neck cancer patient datasets. Our analysis revealed a significant positive correlation between YEATS2 expression and increasing tumor grade. Specifically, we observed significantly higher YEATS2 expression in Grade 2-4 tumors compared to Grade 1 tumors. Given that higher tumor grades are frequently associated with increased metastatic potential and poorer prognosis in HNC[2], this finding provides compelling clinical correlative evidence linking elevated YEATS2 expression to more aggressive disease.

Gene Set Enrichment Analysis (GSEA) for Metastasis Pathways: To further explore the biological processes associated with YEATS2 in a clinical context, we performed GSEA on TCGA HNC patient samples stratified by high versus low YEATS2 expression. This analysis robustly demonstrated a positive enrichment of metastasis-related gene sets in the high YEATS2 expression group, compared to the low YEATS2 group. This strengthens the mechanistic link by showing that pathways associated with metastasis are co-ordinately upregulated when YEATS2 is highly expressed.

These new clinical data provide strong correlative evidence supporting a direct association of YEATS2 with metastasis, building upon our detailed mechanistic dissection of its role in EMT.

(3) There seems to be some discrepancy in the invasion data with BICR10 control cells (Figure 2C). BICR10 control cells with mock plasmids, specifically shControl and pEGFP-C3 show an unclear distinction between invasion capacities. Normally, we would expect the control cells to invade somewhat similarly, in terms of area covered, within the same time interval (24 hours here). But we clearly see more control cells invading when the invasion is done with KD and fewer control cells invading when the invasion is done with OE. Are these just plasmid-specific significant effects on normal cell invasion? This needs to be addressed.

We thank the reviewer for their careful examination of Figure 2C and their insightful observation regarding the appearance of the control cells in relation to the knockdown (Figure 2B) and overexpression (Figure 2C) experiments. We understand how, at first glance, the control invasion levels across these panels might seem disparate.

We wish to clarify that Figure 2B (YEATS2 knockdown) and Figure 2C (YEATS2 overexpression) represent two entirely independent experiments, conducted with distinct experimental conditions and methodologies, as detailed in our Methods section.

Specifically:

Figure 2B (Knockdown): Utilizes lentivirus-mediated transduction for stable shRNA delivery (shControl as control).

Figure 2C (Overexpression): Utilizes transfection with plasmid DNA (pEGFP-C3 as control) via a standard transfection reagent.

These fundamental differences in genetic manipulation methods (transduction vs. transfection), along with potential batch-to-batch variations in reagents or cell passage number at the time of each independent experiment, can indeed lead to variations in absolute basal invasion rates of control cells[3].

Therefore, the invasion capacity of BICR10 control cells in Figure 2B (shControl) should only be compared to the YEATS2 knockdown conditions within that same panel. Similarly, the invasion capacity of control cells in Figure 2C (pEGFP-C3) should only be compared to the YEATS2 overexpression conditions within that specific panel. The crucial finding in each panel lies in the relative change in invasion caused by YEATS2 manipulation (knockdown or overexpression) compared to its respective, concurrently run control.

We have ensured that all statistical analyses (as indicated in the figure legends and methods) were performed by comparing the experimental groups directly to their matched internal controls within each independent experiment. The significant increase in invasion upon YEATS2 overexpression and the significant decrease upon YEATS2 knockdown, relative to their respective controls, are robust and reproducible findings.

(4) In Figure 3G, the Western blot shows an unclear band for YEATS2 in shSP1 cells with YEATS2 overexpression condition. The authors need to clearly identify which band corresponds to YEATS2 in this case.

We thank the reviewer for pointing out the ambiguity in the YEATS2 Western blot for the shSP1 + pEGFP-C3-YEATS2 condition in Figure 3G. We apologize for this lack of clarity. The two bands seen in the shSP1+pEGFP-C3-YEATS2 condition correspond to the endogenous YEATS2 band (lower band) and YEATS2-GFP band (upper band, corresponding to overexpressed YEATS2-GFP fusion protein, which has a higher molecular weight). To avoid confusion, the endogenous band is now highlighted (marked by *) in the lane representing the shSP1+pEGFP-C3-YEATS2 condition. We have also updated the figure legend accordingly.

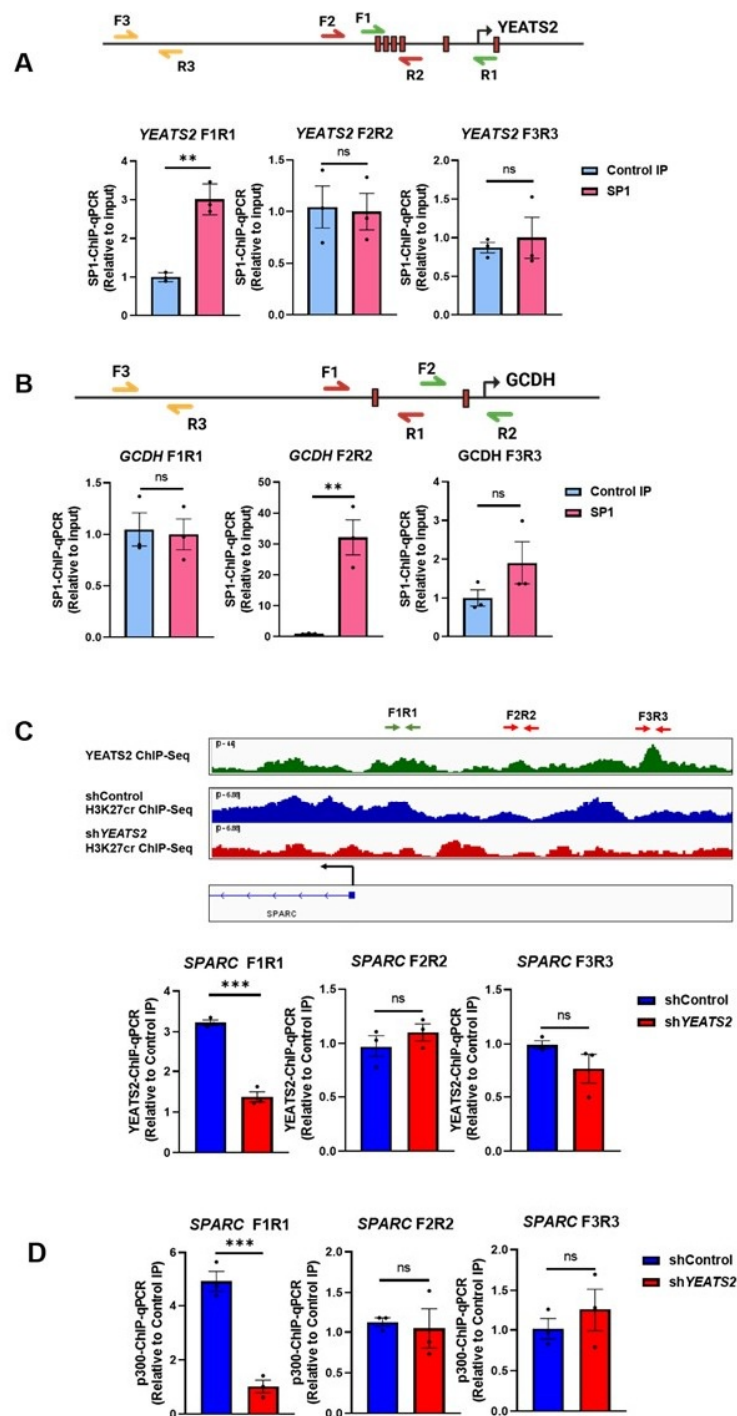
(5) In ChIP assays with SP1, YEATS2 and p300 which promoter regions were selected for the respective genes? Please provide data for all the different promoter regions that must have been analysed, highlighting the region where enrichment/depletion was observed. Including data from negative control regions would improve the validity of the results.

Throughout our study, we have performed ChIP-qPCR assays to check the binding of SP1 on YEATS2 and GCDH promoter, and to check YEATS2 and p300 binding on SPARC promoter. Using transcription factor binding prediction tools and luciferase assays, we selected multiple sites on the YEATS2 and GCDH promoter to check for SP1 binding. The results corresponding to the site that showed significant enrichment were provided in the manuscript. The region of SPARC promoter in YEATS2 and p300 ChIP assay was selected on the basis of YEATS2 enrichment found in the YEATS2 ChIP-seq data. The ChIP-qPCR data for all the promoter regions investigated (including negative controls) can be found below (Author response image 1.).

Authors' response image 1.

(A) SP1 ChIP-qPCR results indicating SP1 occupancy on different regions of YEATS2 promoter. YEATS2 promoter region showing SP1 binding sites (indicated by red boxes) is shown above. SP1 showed significant enrichment at F1R1 region. The results corresponding to F1R1 region were included in Figure 3D. (B) SP1 ChIP-qPCR results indicating SP1 occupancy on different regions of GCDH promoter. GCDH promoter region showing SP1 binding sites (indicated by red boxes) is shown above. SP1 showed significant enrichment at F2R2 region. The results corresponding to F2R2 region were included in Figure 7E. (C) YEATS2 ChIP-qPCR results in shControl vs. shYEATS2 BICR10 cells indicating YEATS2 occupancy on different regions of SPARC promoter. SPARC promoter region showing YEATS2 ChIP-seq and H3K27cr ChIP-seq signals is shown above. YEATS2 showed significant enrichment at F1R1 region. The results corresponding to F1R1 region were included in Figure 5C. (D) p300 ChIP-qPCR results in

shControl vs. shYEATS2 BICR10 cells indicating p300 occupancy on different regions of SPARC promoter. p300 showed significant enrichment at F1R1 region. The results corresponding to F1R1 region were included in Figure 5F.



(6) The authors establish a link between H3K27Cr marks and GCDH expression, and this is an already well-known pathway. A critical missing piece is the level of ECHS1 in patient samples. This will clearly delineate if the balance shifted towards crotonylation.

We greatly appreciate the reviewer's insightful comment regarding the importance of assessing ECHS1 levels in patient samples to clearly delineate the metabolic balance shifting towards crotonylation. We fully agree that this is a critical piece of evidence.

To directly address this point and substantiate our claim regarding the altered metabolic balance in HNC, we had previously analyzed the expression of both GCDH and ECHS1 in TCGA HNC RNA-seq data (as presented in Figure 4—figure supplement 1A and B). This analysis revealed a consistent increase in GCDH expression and a concomitant decrease in ECHS1 expression in tumor samples compared to normal tissues. Based on these findings, we hypothesized that this altered expression profile would indeed lead to an accumulation of crotonyl-CoA and, consequently, an overall increase in histone crotonylation in HNC.

To further validate and extend these findings at the protein level, we have now performed immunohistochemistry (IHC) analysis for both ECHS1 and GCDH in a cohort of HNC normal vs. tumor tissues. Our IHC results strikingly corroborate the RNA-seq data: GCDH consistently showed increased protein expression in tumor samples, whereas ECHS1 exhibited significantly reduced protein expression in tumors compared to their adjacent normal counterpart tissues (Figure 4E and Authors' response figure 5).

These new data, combined with existing TCGA HNC RNA-seq analysis strongly supports our proposed mechanism where altered GCDH and ECHS1 expression contributes to increased histone crotonylation in head and neck cancer.

(7) The p300 ChIP data on the SPARC promoter is confusing. The authors report reduced p300 occupancy in YEATS2-silenced cells, on SPARC promoter. However, this is paradoxical, as p300 is a writer, a histone acetyltransferase (HAT). The absence of a reader (YEATS2) shouldn't affect the writer (p300) unless a complex relationship between p300 and YEATS2 is present. The role of p300 should be further clarified in this case. Additionally, transcriptional regulation of SPARC expression in YEATS2 silenced cells could be analysed via downstream events, like Pol-II recruitment. Assays such as Pol-II ChIP-qPCR could help explain this.

We greatly appreciate the reviewer's insightful observation regarding the apparently paradoxical reduction of p300 occupancy on the SPARC promoter upon YEATS2 silencing (Figure 5F), and their call for further clarification of p300's role and the potential complex relationship with YEATS2. We agree that this point required further mechanistic investigation.

As we have shown through RNA-seq and ChIP-seq analyses, YEATS2 broadly influences histone crotonylation levels at gene promoters, thereby impacting gene expression. While p300 is indeed a known histone acetyltransferase (HAT) with promiscuous acyltransferase activity, including crotonyltransferase activity[4], the precise mechanism by which its occupancy is affected by a 'reader' protein like YEATS2 was unclear. Our initial data suggested a dependency of p300 recruitment on YEATS2.

To directly address the reviewer's concern and thoroughly delineate the molecular mechanism of cooperativity between YEATS2 and p300 in regulating histone crotonylation, we have now performed a series of targeted experiments, which have been incorporated into the revised manuscript:

(a) Validation of p300's role in SPARC expression: We performed p300 knockdown in BICR10 cells, followed by immunoblotting to assess SPARC protein levels. As expected, a significant decrease in SPARC protein levels was observed upon p300 knockdown (Figure 5G). This confirms p300's direct involvement in SPARC gene expression.

(b) Direct interaction between YEATS2 and p300: To investigate a potential physical association, we performed co-immunoprecipitation assays to check for an interaction between endogenous YEATS2 and p300. Our results clearly demonstrate the presence of YEATS2 in the p300-immunoprecipitate sample, indicating that YEATS2 and p300 physically interact and likely function together as a complex to drive the expression of target genes like SPARC (Figure 5H). This direct interaction provides the mechanistic basis for how YEATS2 influences p300 occupancy.

(c) Impact on transcriptional activity (Pol II recruitment): As suggested, we performed RNA Polymerase II (Pol II) ChIP-qPCR on the SPARC promoter in YEATS2 knockdown cells. We observed a significant decrease in Pol II occupancy on the SPARC promoter after YEATS2 knockdown in BICR10 cells (Figure 6C). This confirms that YEATS2 silencing leads to reduced transcriptional initiation/elongation at this promoter.

(d) p300's direct role in H3K27cr on SPARC promoter: To confirm p300's specific role in crotonylation at this locus, we performed H3K27cr ChIP-qPCR after p300 knockdown. As anticipated, a significant decrease in H3K27cr enrichment was observed on the SPARC promoter upon p300 knockdown (Figure 6J), directly demonstrating p300's crotonyltransferase activity at this site.

(e) Rescue of p300 occupancy and H3K27cr by YEATS2 overexpression in SP1deficient cells: To further establish the YEATS2-p300 axis, we performed SP1 knockdown (which reduces YEATS2 expression) followed by ectopic YEATS2 overexpression, and then assessed p300 occupancy and H3K27cr levels on the SPARC promoter. While SP1 knockdown led to a decrease in both p300 and H3K27cr enrichment, we observed a significant rescue of both p300 occupancy and H3K27cr enrichment upon YEATS2 overexpression in the shSP1 cells (Figure 6E and F). This provides strong evidence that YEATS2 acts downstream of SP1 to regulate p300 recruitment and H3K27cr levels.

Collectively, these comprehensive new results clearly establish that YEATS2 directly interacts with and assists in the recruitment of p300 to the SPARC promoter. This recruitment is crucial for p300's localized crotonyltransferase activity, leading to increased H3K27cr marks and subsequent activation of SPARC transcription. This clarifies the previously observed 'paradox' and defines a novel cooperative mechanism between a histone reader (YEATS2) and a writer (p300) in regulating histone crotonylation and gene expression.

(8) The role of GCDH in producing crotonyl-CoA is already well-established in the literature. The authors' hypothesis that GCDH is essential for crotonyl-CoA production has been proven, and it's unclear why this is presented as a novel finding. It has been shown that YEATS2 KD leads to reduced H3K27cr, however, it remains unclear how the reader is affecting crotonylation levels. Are GCDH levels also reduced in the YEATS2 KD condition? Are YEATS2 levels regulating GCDH expression? One possible mechanism is YEATS2 occupancy on GCDH promoter and therefore reduced GCDH levels upon YEATS2 KD. This aspect is crucial to the study's proposed mechanism but is not addressed thoroughly.

We appreciate the reviewer's valuable comment questioning the novelty of GCDH's role in crotonyl-CoA production and seeking further clarification on how YEATS2 influences crotonylation levels beyond its reader function.

We agree that GCDH's general role in producing crotonyl-CoA is well-established[5,6]. Our study, however, aims to delineate a novel epigenetic-metabolic crosstalk in head and neck cancer, specifically investigating how the interplay between the histone crotonylation reader YEATS2 and the metabolic enzyme GCDH contributes to increased histone crotonylation and drives EMT in this context.

Our initial investigations using GSEA on publicly available TCGA RNA-seq data revealed that HNC patients with high YEATS2 expression also exhibit elevated expression of genes involved in the lysine degradation pathway, prominently including GCDH. Recognizing the known roles of YEATS2 in preferentially binding H3K27cr and GCDH in producing crotonylCoA, we hypothesized that the elevated H3K27cr levels observed in HNC are a consequence of the combined action of both YEATS2 and GCDH. We have provided evidence that increased nuclear GCDH correlates with higher H3K27cr abundance, likely due to an increased nuclear pool of crotonyl-CoA, and that YEATS2 contributes through its preferential maintenance of crotonylation marks by recruiting p300 (as detailed in Figure 5FH and Figure 6J-L of the manuscript and elaborated in our response to point 7). Thus, our work highlights that both YEATS2 and GCDH are crucial for the regulation of histone crotonylation-mediated gene expression in HNC.

To directly address the reviewer's query regarding YEATS2's influence on GCDH levels and nuclear histone crotonylation:

- YEATS2 does not transcriptionally regulate GCDH: We did not find any evidence of YEATS2 directly regulating the expression levels of GCDH at the transcriptional level in HNC cells.
- Novel finding: YEATS2 regulates GCDH nuclear localization: Crucially, we discovered that YEATS2 downregulation significantly reduces the nuclear pool of GCDH in head and neck cancer cells (Figure 7G). This is a novel mechanism suggesting that YEATS2 influences histone crotonylation not only by affecting promoter H3K27cr levels via p300 recruitment, but also by regulating the availability of the crotonyl-CoA producing enzyme, GCDH, within the nucleus.
- Common upstream regulation by SP1: Interestingly, we found that both YEATS2 and GCDH expression are commonly regulated by the transcription factor SP1 in HNC. Our data demonstrate that SP1 binds to the promoters of both genes, and its downregulation leads to a decrease in their respective expressions (Figure 3 and Figure 7). This provides an important upstream regulatory link between these two key players.
- Functional validation of GCDH in EMT: We further assessed the functional importance of GCDH in maintaining the EMT phenotype in HNC cells. Matrigel invasion assays after GCDH knockdown and overexpression in BICR10 cells revealed that the invasiveness of HNC cells was significantly reduced upon GCDH knockdown and significantly increased upon GCDH overexpression (results provided in revised manuscript Figure 7F and Figure 7—figure supplement 1F).

These findings collectively demonstrate a multifaceted role for YEATS2 in regulating histone crotonylation by both direct recruitment of the writer p300 and by influencing the nuclear availability of the crotonyl-CoA producing enzyme GCDH. We acknowledge that the precise molecular mechanism governing YEATS2's effect on GCDH nuclear localization remains an exciting open question for future investigation, but our current data establishes a novel regulatory axis.

(9) The authors should provide IHC analysis of YEATS2, SPARC alongside H3K27cr and GCDH staining in normal vs. tumor tissues from HNC patients.

We thank the reviewer for their suggestion. We have performed IHC analysis for YEATS2, H3K27cr and GCDH in normal and tumor samples obtained from HNC patient.

Reviewer #2 (Public review):

Summary:

The manuscript emphasises the increased invasive potential of histone reader YEATS2 in an SP1-dependent manner. They report that YEATS2 maintains high H3K27cr levels at the promoter of EMT-promoting gene SPARC. These findings assigned a novel functional implication of histone acylation, crotonylation.

We thank the reviewer for the constructive comments. We are committed to making beneficial changes to the manuscript in order to alleviate the reviewer's concerns.

Concerns:

(1) The patient cohort is very small with just 10 patients. To establish a significant result the cohort size should be increased.

We thank the reviewer for this suggestion. We have increased the number of patient samples to assess the levels of YEATS2 (n=23 samples) and the results have been included in Figure 1G and Figure 1—figure supplement 1F.

(2) Figure 4D compares H3K27Cr levels in tumor and normal tissue samples. Figure 1G shows overexpression of YEATS2 in a tumor as compared to normal samples. The loading control is missing in both. Loading control is essential to eliminate any disparity in protein concentration that is loaded.

To address the reviewer's concern, we have repeated the experiment and used H3 as a loading control as nuclear protein lysates from patient samples were used to check YEATS2 and H3K27cr levels.

(3) Figure 4D only mentions 5 patient samples checked for the increased levels of crotonylation and hence forms the basis of their hypothesis (increased crotonylation in a tumor as compared to normal). The sample size should be more and patient details should be mentioned.

As part of the revision, we have now checked the H3K27cr levels in a total of 23 patient samples and the results have been included in Figure 4D and Figure 4—figure supplement 1D. Patient details are provided in Supplementary Table 6.

(4) YEATS2 maintains H3K27Cr levels at the SPARC promoter. The p300 is reported to be hyper-activated (hyperautoacetylated) in oral cancer. Probably, the activated p300 causes hyper-crotonylation, and other protein factors cause the functional translation of this modification. The authors need to clarify this with a suitable experiment.

We thank the reviewer for this insightful comment regarding the functional relationship between YEATS2 and p300 in the context of H3K27cr, especially considering reports of p300 hyper-activation in oral cancer. We agree that a precise clarification of p300's role and its cooperativity with YEATS2 is crucial to fully understand the functional translation of this modification.

As we have shown through global RNA-seq and ChIP-seq analyses, YEATS2 broadly affects gene expression by regulating histone crotonylation levels at gene promoters. We also recognize that the histone writer p300 is a promiscuous acyltransferase, known to add various non-acetyl marks, including crotonylation[4]. Our initial data, showing decreased

p300 occupancy on the *SPARC* promoter upon YEATS2 downregulation (Figure 5F), suggested a strong dependency of p300 on YEATS2 for its recruitment. To fully delineate the molecular mechanism of this cooperativity and clarify how YEATS2 influences p300-mediated histone crotonylation and its functional outcomes, we have performed the following series of experiments, which have been integrated into the revised manuscript:

(a) Validation of p300's role in *SPARC* expression: We performed p300 knockdown in BICR10 cells, followed by immunoblotting to assess *SPARC* protein levels. As expected, a significant decrease in *SPARC* protein levels was observed upon p300 knockdown (Figure 5G). This confirms p300's direct involvement in *SPARC* gene expression.

(b) Direct interaction between YEATS2 and p300: To investigate a potential physical association, we performed co-immunoprecipitation assays to check for an interaction between endogenous YEATS2 and p300. Our results clearly demonstrate the presence of YEATS2 in the p300-immunoprecipitate sample, indicating that YEATS2 and p300 physically interact and likely function together as a complex to drive the expression of target genes like *SPARC* (Figure 5H). This direct interaction provides the mechanistic basis for how YEATS2 influences p300 occupancy.

(c) Impact on transcriptional activity (Pol II recruitment): As suggested, we performed RNA Polymerase II (Pol II) ChIP-qPCR on the *SPARC* promoter in YEATS2 knockdown cells. We observed a significant decrease in Pol II occupancy on the *SPARC* promoter after YEATS2 knockdown in BICR10 cells (Figure 6C). This confirms that YEATS2 silencing leads to reduced transcriptional initiation/elongation at this promoter.

(d) p300's direct role in H3K27cr on *SPARC* promoter: To confirm p300's specific role in crotonylation at this locus, we performed H3K27cr ChIP-qPCR after p300 knockdown. As anticipated, a significant decrease in H3K27cr enrichment was observed on the *SPARC* promoter upon p300 knockdown (Figure 6J), directly demonstrating p300's crotonyltransferase activity at this site.

(e) Rescue of p300 occupancy and H3K27cr by YEATS2 overexpression in SP1deficient cells: To further establish the YEATS2-p300 axis, we performed SP1 knockdown (which reduces YEATS2 expression) followed by ectopic YEATS2 overexpression, and then assessed p300 occupancy and H3K27cr levels on the *SPARC* promoter. While SP1 knockdown led to a decrease in both p300 and H3K27cr enrichment, we observed a significant rescue of both p300 occupancy and H3K27cr enrichment upon YEATS2 overexpression in the sh_SP1_ cells (Figure 6K and L). This provides strong evidence that YEATS2 acts downstream of SP1 to regulate p300 recruitment and H3K27cr levels.

Collectively, these comprehensive new results clearly establish that YEATS2 directly interacts with and assists in the recruitment of p300 to the *SPARC* promoter. This recruitment is crucial for p300's localized crotonyltransferase activity, leading to increased H3K27cr marks and subsequent activation of *SPARC* transcription. This clarifies the previously observed 'paradox' and defines a novel cooperative mechanism between a histone reader (YEATS2) and a writer (p300) in regulating histone crotonylation and gene expression.

(5) *I do not entirely agree with using GAPDH as a control in the western blot experiment since GAPDH has been reported to be overexpressed in oral cancer.*

We would like to clarify that GAPDH was not used as a loading control for protein expression comparisons between normal and tumor samples. GAPDH was used as a loading control only in experiments using head and neck cancer cell lines where shRNA-mediated knockdown or overexpression was employed. These manipulations specifically target the genes of interest and are not expected to alter GAPDH expression, making it a suitable loading control in these instances.

(6) The expression of EMT markers has been checked in shControl and shYEATS2 transfected cell lines (Figure 2A). However, their expression should first be checked directly in the patients' normal vs. tumor samples.

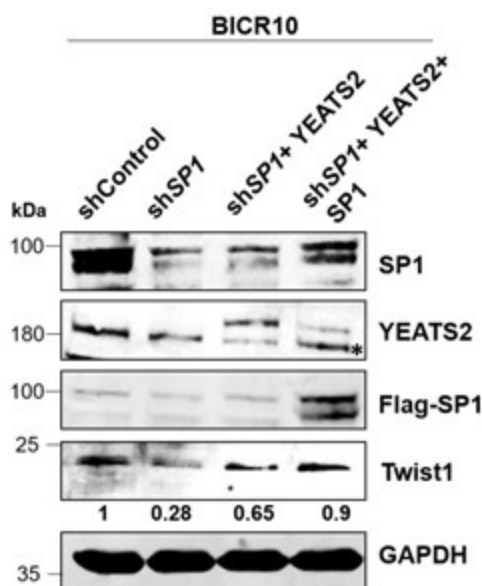
We thank the reviewer for the suggestion. We have now checked the expression of EMT marker Twist1 alongside YEATS2 expression in normal vs. tumor tissue samples using IHC (Figure 4E).

(7) In Figure 3G, knockdown of SP1 led to the reduced expression of YEATS2 controlled gene Twist1. Ectopic expression of YEATS2 was able to rescue Twist1 partially. In order to establish that SP1 directly regulates YEATS2, SP1 should also be re-introduced upon the knockdown background along with YEATS2 for complete rescue of Twist1 expression.

To address the reviewer's concern regarding the partial rescue of Twist1 in SP1 depleted-YEATS2 overexpressed cells, we performed the experiment as suggested by the reviewer. We overexpressed both SP1 and YEATS2 in SP1-depleted cells and found that Twist1 depletion was almost completely rescued.

Authors' response image 2.

Immunoblot depicting the decreased Twist1 levels on SP1 knockdown and its subsequent rescue of expression upon YEATS2 and SP1 overexpression in BICR10 (endogenous YEATS2 band indicated by *).



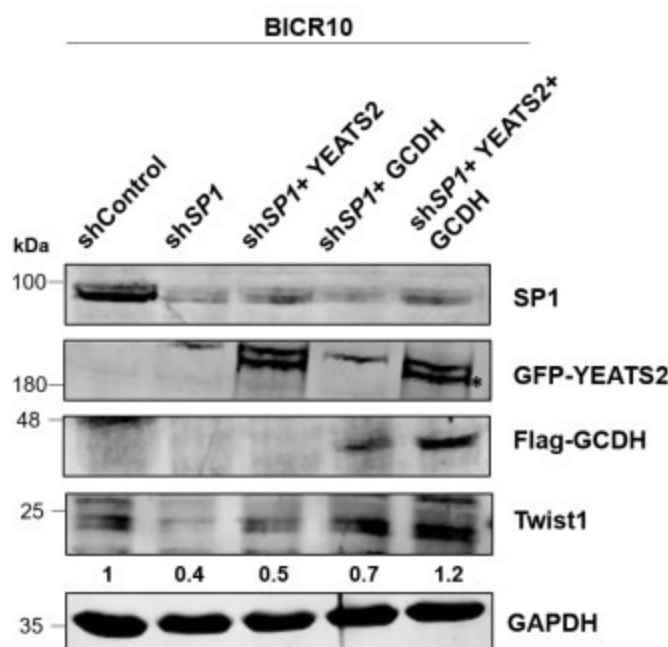
(8) In Figure 7G, the expression of EMT genes should also be checked upon rescue of SPARC expression.

We thank the reviewer for the suggestion. We have examined the expression of EMT marker Twist1 on YEATS2/ GCDH rescue. On overexpressing both YEATS2 and GCDH in sh_SP1_ cells we found that the depleted expression of Twist1 was rescued.

Authors' response image 3.

Immunoblot depicting the decreased Twist1 levels on SP1 knockdown and its subsequent rescue of expression upon dual overexpression of YEATS2 and GCDH in BICR10 (* indicates

GFP-tagged YEATS2 probed using GFP antibody).



Reviewer #1 (Recommendations for the authors):

While the study offers insights into the specific role of this axis in regulating epithelial-to-mesenchymal transition (EMT) in HNC, its broader mechanistic novelty is limited by prior discoveries in other cancer types (<https://doi.org/10.1038/s41586-023-06061-0>). The manuscript would benefit from the inclusion of metastasis data, the role of key metabolic enzymes like ECHS1, the molecular mechanisms governing p300 and YEATS2 interactions, additional IHC data, negative control data in ChIP, and an explanation of discrepancies in certain figures.

We thank the reviewer for their constructive suggestions. We have made extensive revisions to our manuscript to substantiate our findings. We have looked into the expression of ECHS1/GCDH in HNC tumor tissues using IHC, performed extensive experiments to validate the role of p300 in YEATS2-mediated histone crotonylation, and provided additional data supporting our findings wherever required. The revised figures have been provided in the updated version of the manuscript and also in the Authors' response.

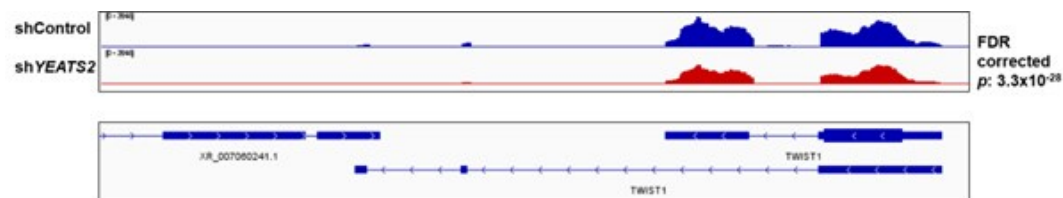
Minor Comments:

(1) The study begins with a few EMT markers, such as Vimentin, Twist, and N-Cadherin to validate the role of YEATS2 in promoting EMT. Including a broader panel of EMT markers would strengthen the conclusions about the effects of YEATS2 on EMT and invasion. Additionally, the rationale for selecting these EMT markers is not fully elaborated. Why were other well-known EMT players not included in the analysis?

On performing RNA-seq with shControl and sh_YEATS2_ samples, we discovered that *TWIST1* was showing decrease in expression on YEATS2 downregulation. So Twist1 was investigated as a potential target of YEATS2 in HNC cells. N-Cadherin was chosen because it is known to get upregulated directly by Twist1[8]. Further, Vimentin was chosen as it a well-known marker for mesenchymal phenotype and is frequently used to indicate EMT in cancer cells[9].

Authors' response image 4.

IGV plot showing the decrease in Twist1 expression in shControl vs. shYEATS2 RNA-seq data.



Other than the EMT-markers used in our study, the following markers were amongst those that showed significant change in gene expression on YEATS2 downregulation.

Authors' response table 1.

List of EMT-related genes that showed significant change in expression on YEATS2 knockdown in RNA-seq analysis.

HUGO Gene Symbol	Common Name	Marker Type	Status in shControl vs. shYEATS2 RNA-seq
COL4A1	Collagen IV alpha 1	Epithelial	Downregulated
FN1	Fibronectin	Mesenchymal	Downregulated
S100A4	S100A4	Mesenchymal	Downregulated
CTNNB1	Beta-catenin	Dual	Downregulated
CDH11	Cadherin-11	Dual	Downregulated
MMP2	MMP2	Mesenchymal	Downregulated
TWIST1	Twist1	Mesenchymal	Downregulated
CDH1	E-cadherin	Epithelial	Upregulated
GSK3B	GSK3B	Dual	Upregulated
SDC1	Syndecan-1	Epithelial	Upregulated
CLDN1	Claudin-1	Epithelial	Upregulated

As depicted in the table above, majority of the genes that showed downregulation on YEATS2 knockdown were mesenchymal markers, while epithelial-specific genes such as Ecadherin and Claudin-1 showed upregulation. This data signifies the essential role of YEATS2 in driving EMT in head and neck cancer.

(2) The authors use Ponceau staining, but the rationale behind this choice is unclear. Ponceau is typically used for transfer validation. For the same patient, western blot loading controls like Actin/GAPDH should be shown. Also, at various places throughout the manuscript, Ponceau staining has been used. These should also be replaced with Actin/GAPDH blots.

Ponceau S staining is frequently used as alternative for housekeeping genes like GAPDH as control for protein loading[10]. However, to address this issue, we have repeated the western

and used H3 as a loading control as nuclear protein lysates from patient samples were used to check YEATS2 and H3K27cr levels.

For experiments (In Figures 5E, 6F, 6I, and 7H) where we assessed SPARC levels in conditioned media obtained from BICR10 cells (secretory fraction), Ponceau S staining was deliberately used as the loading control. In such extracellular protein analyses, traditional intracellular housekeeping genes (like Actin or GAPDH) are not applicable. Ponceau S has been used as a control for showing SPARC expression in secretory fraction of mammalian cell lines in previous studies as well¹¹.

(3) The manuscript briefly mentions that p300 was identified as the only protein with increased expression in tumours compared to normal tissue in the TCGA dataset. What other writers were checked for? Did the authors check for their levels in HNC patients?

We thank the reviewer for this observation. As stated by previous studies [12,13], p300 and GCN5 are the histone writers that can act as crotonyltransferases at the H3K27 position. Although the crotonyltransferase activity of GCN5 has been demonstrated in yeast, it has not been confirmed in human. Whereas the histone crotonyltransferase activity of p300 has been validated in human cells using in vitro HCT assays[4,14]. Therefore, we chose to focus on p300 for further validation of its role in YEATS2mediated regulation of histone crotonylation. We did not check the levels of p300 in HNC patient tissues. However, p300 showed higher expression in tumor as compared to normal in publicly available HNC TCGA RNA-seq data (Figure 5—figure supplement 1G).

We acknowledge that the original statement in the manuscript, 'For this we looked at expression of the known writers of H3K27Cr mark in TCGA dataset, and discovered that p300 was the only protein that had increased expression in tumor vs. normal HNC dataset...', was indeed slightly misleading. Our intention was to convey that p300 is considered the major and most validated histone crotonyltransferase capable of influencing crotonylation at the H3K27 position in humans, and that its expression was notably increased in the HNC TCGA tumor dataset. We have now reframed this sentence in the revised manuscript to accurately reflect our findings and focus, as follows:

'For this, we checked the expression of p300, a known writer of H3K27cr mark in humans, in the TCGA dataset. We found that p300 had increased expression in tumor vs. normal HNC dataset...'

This revised wording more accurately reflects our specific focus on p300's established role and its observed upregulation in HNC.

(4) Figure 6E, blot should be replaced. The results aren't clearly visible.

We thank the reviewer for this observation. We have repeated the western blot and the Figure 6E (Figure 6F in the revised version of manuscript) has now been replaced with a cleaner blot.

(5) Reference 9 and 19 are the same. Please rectify.

We apologize for this inadvertent error. We have rectified this error in the updated version of the manuscript.

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<https://doi.org/10.7554/eLife.103321.2.sa0>